The market for voluntary carbon offsets *

Florian Berg[1], Marco Ceccarelli[2], Florian Heeb[3,1], Alexey Ivashchenko[†2], Roberto Rigobon[1], and Remco C.J. Zwinkels[2,4]

[1]MIT - Sloan School of Management
[2]VU Amsterdam
[3]Leibniz Institute for Financial Research SAFE / Goethe University Frankfurt
[4]Tinbergen Institute

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Abstract

We study pricing in the voluntary carbon market (VCM) using a proprietary transaction-level dataset from a leading dealer. Prices are highly dispersed, from cents to over $100 per ton of CO_2. Credits from less reliable technologies, but with positive non-carbon externalities, command prices two to four times those of trusted technologies. Within the same offset project, buyer identity explains substantially more price variation than in other OTC markets. High-volume buyers pay less; heavy polluters and firms with sustainability commitments do not pay more. Our findings suggest the VCM currently does not provide a reliable market-based carbon price signal.

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[†]Corresponding author. Mail:Alexey.Ivashchenko@vu.nl

1 Introduction

While the prospect of a compulsory global carbon pricing mechanism remains elusive, the voluntary carbon market (VCM) has emerged as a private market-based substitute. Firms participate in the VCM to manage their carbon footprints and shape how investors and clients perceive their environmental performance. Within this market, companies purchase carbon credits issued by projects that claim to prevent the release of one metric ton of CO_2 or to remove an equivalent amount from the atmosphere. Despite growing criticism about the environmental integrity of many such projects, the supply of carbon credits continues to expand. Yet, the economics of the VCM remain poorly understood, in part due to the opacity of offset projects and, especially, its over-the-counter pricing structure. Indeed, the fundamental nature of the VCM itself remains unclear: does it function as a global, market-based carbon price signal, or do carbon credit prices primarily capture buyer preferences and incentives, with genuine emission reductions playing only a secondary role?

We address this question with a novel proprietary dataset from one of the world's largest VCM dealers, covering about 11% of secondary VCM transactions by dollar value from 2018 to 2024. In this first-of-its-kind dataset, we are able to observe both anonymized buyer and project characteristics.[1] We supplement this with an independent transaction sample from a major exchange and a commodity intelligence provider, which we link to external project-quality and co-benefit measures.

Two pricing facts stand out. Across projects, transaction prices vary by more than two orders of magnitude—from a few cents to over one hundred dollars per ton of CO_2 (Figure 1). Even within projects, the same credits can simultaneously trade at prices differing by a factor of two or more.

[1] The existing literature examining VCM relies instead on third-party pricing information that reflects either average transaction prices, bid-ask midpoints in the absence of trading, or estimated prices (e.g., Conte and Kotchen 2010, Kim et al. 2024, or Pedraza et al. 2026). Given the large within-project price dispersion, observing actual transaction prices is a key advantage of our data.

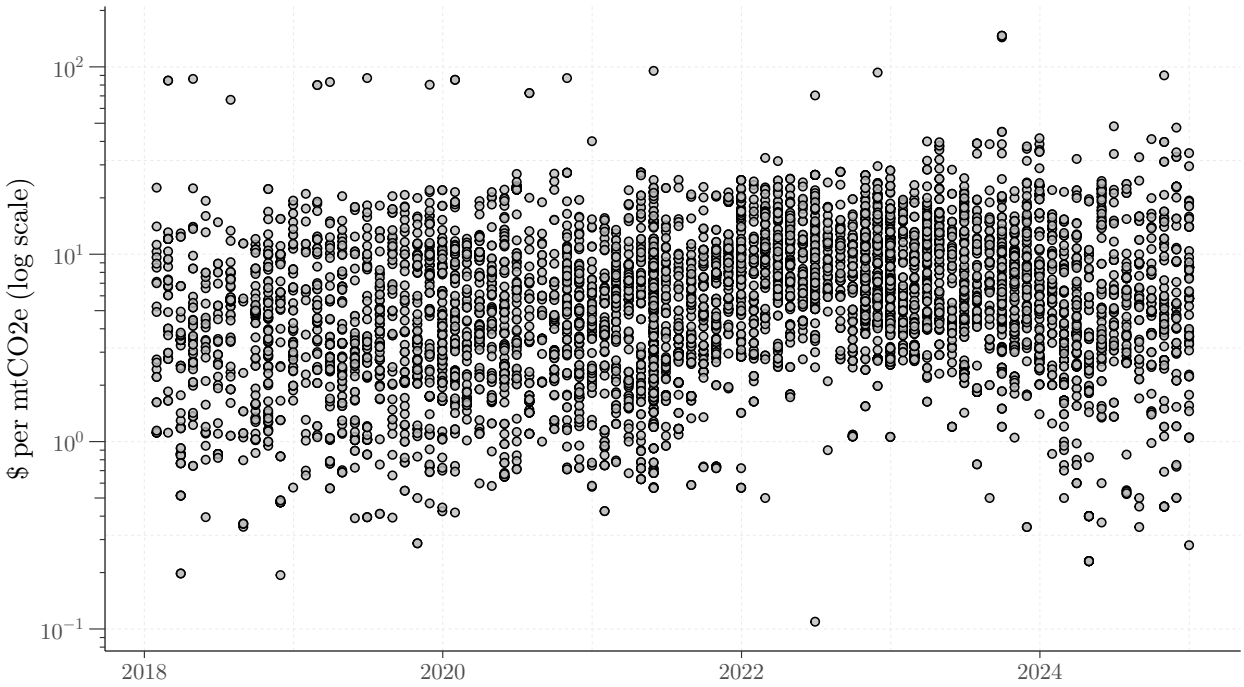


Figure 1: **Sales prices in the secondary market for voluntary carbon credits.**
Each point represents a secondary market transaction between a large voluntary credits dealer (seller) and a non-dealer customer (buyer). The transaction date is on the x-axis. The transaction price, in $ per metric ton of CO_2 equivalent, is on the y-axis (log-scale).

This suggests that VCM credits are priced as heterogeneous goods, not as a homogeneous commodity in which one ton of CO_2 abated has a single market price. To understand the sources of this heterogeneity, we estimate hedonic pricing models that relate transaction prices to project, transaction, and buyer characteristics.

Project technology is the largest single project-level price factor, explaining about 29% of price variation. Credits from nature-based projects that target carbon removal or emission reductions, and from household-level interventions, trade at premiums of 99–362% over renewable energy credits, while credits from waste management and industrial process projects trade at no significant premium. Among the technologies for which external climate-effectiveness assessments exist, premia and reliability are decoupled: nature-based avoidance and household credits, both rated low

on delivery of promised emission reductions and on additionality[2] (Probst et al., 2024; Cames et al., 2017), trade well above renewables, while waste and industrial credits (with stronger external assessments) trade at no premium.[3] The nature-based avoidance premium does not decline following a salient 2023 *New Yorker* investigation that documented serious quality concerns about a large-scale project.

Evidence on project-level indicators is mixed but points to a pronounced role for non-carbon benefits. Credit-reliability ratings from BeZero Carbon, observable in an auxiliary sample, correlate positively with prices unconditionally but lose significance once accounting for country fixed effects. To test for the value buyers place on non-carbon co-benefits, we count self-reported Sustainable Development Goal (SDG) contributions for projects. This measure correlates positively with prices but weakens when country fixed effects are included. Conditional on country fixed effects, credits certified by Gold Standard, a registry whose methodology emphasizes co-benefits, trade at a premium of up to 36%.

Buyer-side price variation is substantial, even after accounting for project characteristics. Financial and consumer-sector buyers pay 9–22% more than industrial manufacturing buyers; larger firms pay about 6% more than small and medium-sized firms; top-tier clients pay 16–23% less than other buyers; and a 10% increase in buyers' country GDP per capita correlates with 0.9–1.3% higher prices. Other observable characteristics do not predict higher prices: buyers from heavy-polluting industries (energy and transportation), firms publicly committed to long-term sustainability targets (SBTi signatories), and firms domiciled in highly polluting countries do not pay more for VCM credits. Buyer-side factors explain a substantial share of price variation: buyer identities add 21% of explanatory power on top of all observable project, transaction, and buyer factors, and a regres-

[2] In the VCM context, projects are referred to as additional if they were enabled by carbon credit revenues and thus would not have been implemented otherwise.

[3] Nature-based removals carry the largest premia in our data but are not covered by these external benchmarks.

sion of prices on buyer characteristics and buyer IDs alone reaches 62% adjusted R^2. Both shares are orders of magnitude larger than in U.S. corporate bonds, an OTC market where dealer market power is well-documented.[4]

Two further VCM price patterns are common to OTC markets with limited pre-trade price transparency. Holding client identity constant, transaction prices are about 0.5% lower for a 10% increase in trading volume, consistent with dealers offering price concessions to hold less inventory. Non-standard-currency transactions trade at a 26–43% premium, reflecting foreign-exchange risk borne by the dealer.

The introduction of standardized exchange-traded products has been unsuccessful amid substantial price dispersion across VCM projects. The Chicago Mercantile Exchange and Xpansiv CBL launched VCM spot and futures contracts on widely defined baskets of credits during our sample period, but their prices quickly raced to the bottom of the OTC distribution—consistent with the cheapest-to-deliver principle—and both prices and trading volumes collapsed to near zero by 2024.

Together, these patterns are difficult to reconcile with carbon credits trading as a homogeneous commodity. The misalignment between cross-project prices and external reliability measures rules out delivery risk as the dominant pricing factor; instead, the pattern is consistent with buyers valuing non-carbon project attributes such as co-benefits. The large within-project price variation, captured almost entirely by buyer fixed effects, is consistent with intermediary search frictions and dealer price discrimination in this opaque OTC market.

Our findings suggest that VCM credit prices do not function as a market-based carbon price signal but instead reflect heterogeneous buyer valuations of project characteristics beyond carbon reductions and the structure of dealer-intermediated OTC trading. This has important implications

[4]For reference, we estimate the same metric for U.S. corporate bonds, and find that the buyer identities do not explain *any* variation in bond transaction prices beyond what bond price factors already explain.

for policymakers. The VCM could mobilize private capital across jurisdictions without requiring revenue-sharing or centralized authority. To unlock this potential, however, the climate claims firms are permitted to make based on carbon credits would need to be more closely aligned with actual climate outcomes. In addition, pre-trade transparency would need to improve in order to reduce dealer price discrimination.

Related literature. Our results contribute to the emerging literature on the economics of voluntary carbon markets. First, we add to studies examining the demand for carbon credits. Kim et al. (2024) show that, following an exogenous decline in ESG scores, firms offset a larger share of their emissions, suggesting that carbon credits are used to manage investors' perceptions of corporate sustainability. Rodemeier (2025) investigates consumers' willingness to pay for carbon offsetting in a large-scale field experiment and finds that this willingness is price-elastic but does not respond to the stated climate impact of the offset. Bellon et al. (2026) study the use of compliance carbon credits within cap-and-trade systems. They show that firms subject to cap-and-trade systems sort into offset use according to their carbon productivity: more carbon-efficient firms rely primarily on cap-and-trade allowances, whereas less carbon-efficient firms use cheaper offsets carrying invalidation risk. Unlike previous work, we observe actual prices at the project-by-buyer level. This is important, since there is substantial buyer-side heterogeneity in VCM transaction prices that likely reflects both non-carbon attribute valuations and dealer-intermediated price discrimination. This implies that VCM prices do not provide a reliable market-based carbon price signal.

Second, we contribute to research evaluating the climate effectiveness of carbon offset projects. Calel et al. (2025) show that most wind projects financed through carbon credits would likely have been implemented even without such support. In a meta-study, Probst et al. (2024) find that less than 20% of credits across major project types—such as renewable energy, avoided deforestation,

and household-level interventions—reflect real emission reductions. We build on these insights by demonstrating that carbon credit prices for several of these less reliable project types are nonetheless substantially higher than those for more credible alternatives.

Third, our analysis complements the literature on compliance carbon markets, which examines the effectiveness of mandatory emission trading systems in reducing global greenhouse gas emissions. Colmer et al. (2024) find that the EU Emissions Trading System has led to substantial emission reductions without dampening economic activity or inducing offshoring. In contrast, Bartram et al. (2022) show that, under California's cap-and-trade system, financially constrained firms shift production to other states, whereas unconstrained firms do not adjust their emissions. Akey et al. (2024) argue that mandatory emission trading systems may crowd out voluntary emission reduction initiatives. Laurs et al. (2025) find that non-compliance with the EU ETS leads to substantial price shocks and corresponding hikes in the cost of capital. Finally, Shapiro and Walker (2025) shows that prices in compliance-based environmental offset markets provide valuable information about the marginal cost of pollution reduction. We contribute to this literature by showing that the corresponding informativeness of VCM prices is substantially lower, which implies that, under a potential coupling of the VCM with regional compliance schemes, cheapest-to-deliver carbon credits may dilute regional carbon prices.

2 The voluntary carbon market

This section first offers a perspective on why voluntary carbon markets, in principle, could be a useful tool for mitigating climate externalities. Then, it presents the institutional structure of how the market works, from origination to trading and, finally, to the retirement of carbon credits.

2.1 Voluntary carbon markets as a climate policy tool

To put the VCM in perspective, it is useful to consider the existing range of tools employed to curb negative externalities. Direct regulation, or "command and control" tools, has been the response to externalities since at least the Roman Empire. These policies rely on top-down rules and remain foundational in environmental law today.[5] Pigou (1920) proposes that governments ought to instead impose corrective taxes equal to the marginal external damage when private actions generate social costs. Coase (1960) shows that when property rights are clearly defined, and transaction costs are low, affected parties can bargain to efficient outcomes regardless of who holds the rights, without the need for government intervention.[6] This insight inspires both tradable-rights regimes and tort liability approaches. Ostrom (1990, 2005) document how communities self-organize to manage common-pool resources without markets or central regulation, often through polycentric arrangements that combine local, regional, and national institutions. Late-20th-century policy introduced market-based mechanisms, primarily cap-and-trade systems.[7] By setting an aggregate cap and letting firms trade permits, markets can allocate reductions to the lowest-cost abaters. Each of these governance mechanisms is effective within clearly defined jurisdictions. Their extension to global externalities, such as CO_2 emissions, poses coordination and redistribution problems.[8]

The VCM is appealing in this context because it mobilizes global private capital without requiring prior agreement on revenue sharing or centralized authority. Voluntary carbon markets then emerge alongside compliance markets, allowing firms and individuals to purchase credits outside regulatory mandates to offset emissions. They combine Coasian exchange, Pigouvian pricing logic, and Ostrom-style institutional design: standards bodies such as Verra and Gold Standard define

[5] For historical references see Mommsen and Krueger (1985); Finer (1997); and Rosen (2015).
[6] Also see Stigler (1966).
[7] The idea was argued earlier and is attributed to Dales (1968).
[8] To tackle global externalities, Nordhaus (2015) proposes forming 'clubs' of countries committed to tackling climate change, combined with border taxes for goods imported from outside the member countries. However, this approach still faces significant coordination and redistribution challenges.

methodologies and verification procedures. Tied to corporate ESG and net-zero commitments, the VCM channels private capital toward emission-reduction projects in jurisdictions where binding regulation is missing.[9] However, our results show that the market does not currently price carbon as a commodity: prices reflect non-carbon project attributes and buyer-specific willingness to pay rather than the value of avoided emissions. The VCM's potential to fulfill this governance role, therefore, depends on aligning the climate claims firms can make based on credit usage with the credits' actual climate impact.

2.2 Institutional structure of the VCM

How does the market for carbon offsets work? Carbon credits are tradable claims to a reduction or removal of greenhouse gas emissions, typically denominated in metric tons of CO_2 equivalent ($mtCO_2e$). The primary buyers are corporations seeking to offset their emissions; individuals also buy credits, though they account for a small share of the market.

Offset projects span six broad technology categories. *Renewable energy* projects generate electricity from onshore wind, solar, and hydropower in place of fossil fuels. *Nature-based avoidance* projects protect existing forests from deforestation, principally under the REDD+ framework.[10] *Nature-based removal* projects sequester atmospheric carbon through afforestation, reforestation, and revegetation. *Household* projects, concentrated in lower-income countries, primarily distribute improved cookstoves and water-filtering technologies. *Waste* projects capture methane and other emissions from landfills, composting, manure management, and industrial wastewater treatment. *Industrial* projects recover waste energy and abate emissions in transportation, chemical processes, and—more recently—industrial carbon capture and storage.

[9] See MacKenzie (2009); Gold Standard Foundation (2022); and World Bank (2023).

[10] REDD+ (Reducing Emissions from Deforestation and Forest Degradation) is a UNFCCC framework under which developing countries earn payments for avoided deforestation and sustainable forest management; it provides the methodological basis for most nature-based avoidance credits issued in the VCM.

The issuance of carbon credits is governed by voluntary standard-setting bodies, principally Verra and Gold Standard, which operate registries that record all credit transactions. Most standards require that projects apply approved methodologies and undergo independent third-party verification. Quality requirements typically include standardized measurement of climate outcomes; *additionality*, meaning that the emission reduction would not have occurred absent the project; *avoidance of leakage*, meaning that emissions are not displaced elsewhere; and, for removal projects, *permanence*, meaning that the carbon is durably stored. Projects often advertise co-benefits such as biodiversity gains for afforestation initiatives or respiratory-health gains from cleaner-burning cookstoves, and several standard-setters offer separate certification of these co-benefits.

Since its inception in the early 2000s, the VCM has taken the appearance of a traditional commodity market. Project owners issue credits through one of the standard-setting bodies, with each credit assigned a "vintage" year corresponding to the period in which the claimed emission reduction occurred. Credits are typically sold to buyers through specialized intermediaries; although structured exchanges exist, most transactions occur over the counter. Once a buyer uses a credit to substantiate an environmental claim, the credit is "retired"—marked as permanently removed from circulation in the corresponding registry.

VCM trading has experienced a marked expansion and recent contraction, against a longer-term backdrop of sustained growth in credit issuance and retirements. Cumulative VCM transactions reached approximately \$11 bln between 2005 and 2024, corresponding to around 2.5 bln tons of CO_2e (Ecosystem Marketplace, 2025). Both annual issuances and retirements have grown substantially since 2005, reaching roughly 300 and 200 mln mtCO_2e in 2024, respectively (Panels C and D in Figure 2). Issuances have, however, consistently outpaced retirements, leading to a substantial accumulation of unretired credits in the market. Panel A shows that annual trading volumes were relatively stable at 50–150 mln mtCO_2e from 2007 to 2019, rose sharply through 2020, and peaked in

2021 at 493 mln $mtCO_2e$ (approximately $2.1 bln). Volumes have contracted sharply since, falling to 84 mln $mtCO_2e$ in 2024—similar to pre-peak (2007–2019) levels. The contraction in trading coincided with growing public concerns about the integrity of widely traded credit categories.

Both academic and journalistic work substantiate such integrity concerns, documenting that many project categories fall well short of their emission reduction claims (Probst et al., 2024; Calel et al., 2025). Two main private-sector responses have emerged. The Integrity Council for the Voluntary Carbon Market (ICVCM), launched in 2021, published the Core Carbon Principles in 2023 as a cross-standard framework intended to identify projects that reliably deliver the claimed climate outcomes. Independent rating agencies—BeZero Carbon, Sylvera, and Calyx Global—now also provide project-level quality assessments.

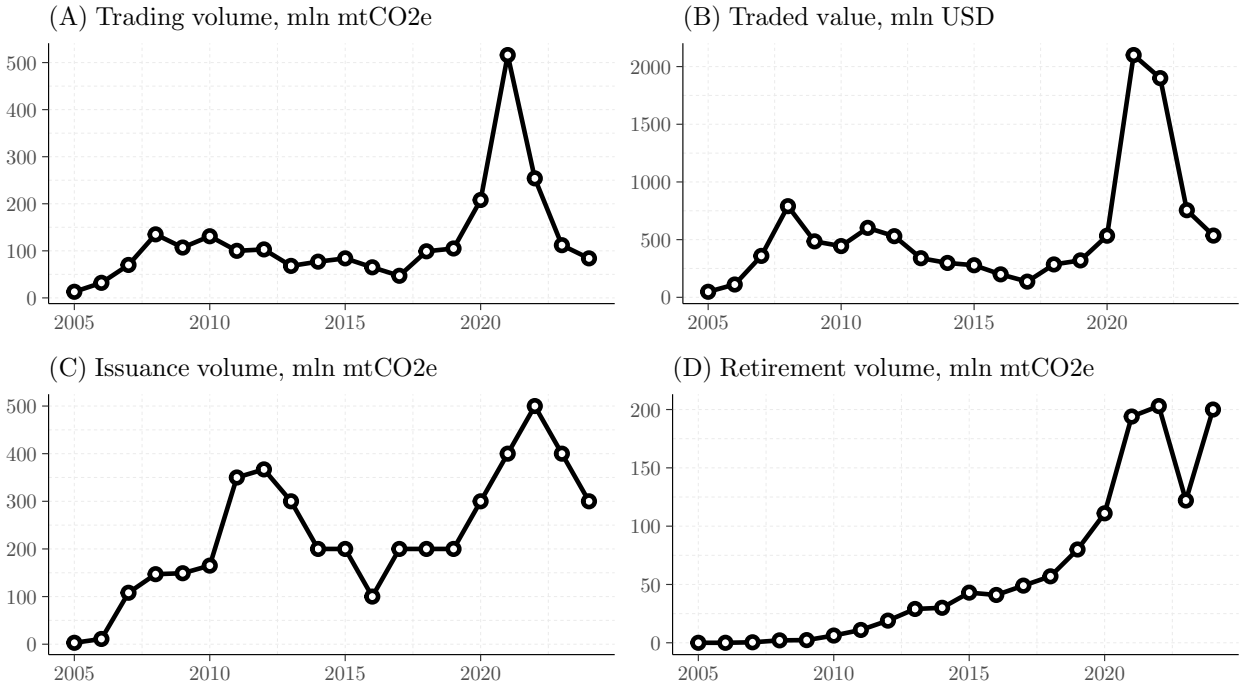


Figure 2: **VCM market size.**
This plot shows aggregate annual market statistics on the VCM. The data are from the Ecosystem Marketplace and refer to global trading in existing voluntary carbon credits (panels A and B), issuance of new credits (panel C), and retirement of existing credits (panel D). 'Mln $mtCO_2e$' stands for million metric tons of CO_2 equivalent.

3 Data

We next introduce our proprietary VCM transaction dataset, describing the data source and sample construction, reporting sample composition and summary statistics, and benchmarking representativeness against an independent transaction sample.

3.1 Data source and sample construction

Our dataset is provided by one of the world's largest VCM broker-dealers and comprises over-the-counter (OTC) carbon credit sales by the dealer to its customers between 2018 and 2024.[11] For each transaction, we observe trade characteristics together with project and customer characteristics. Project and customer identities are anonymized but remain constant throughout the dataset, allowing us to track different trades by the same customer and different transactions of the same project. This is an advantage relative to existing commercial data providers that either estimate average prices across all transactions within a project or, in the absence of trading, report midpoints of indicative (non-executable) bid and ask prices from illiquid trading platforms.

Before applying any filters, the data contain transactions totaling almost $700 mln, approximately 11% of the total VCM secondary market transaction volume over the same period (Ecosystem Marketplace, 2025). We apply four filters to construct the sample. First, we remove transactions with delivery nine months or more after the signing date, about a quarter of the original sample.[12] Second, we drop transactions with prices below 10¢ or above $250 per $mtCO_2e$ (close to the 0.5^{th} and 99.5^{th} percentiles of transaction prices). Third, we exclude credits eligible for local compliance markets in Australia, Colombia, and Switzerland (about 12% of the original sample), since they

[11] For a study of project-level prices in the early days of the VCM in 2008, see Conte and Kotchen (2010).

[12] We aim to analyze spot or short-term future carbon prices rather than describe their full term structure. There are no standard delivery terms in the VCM, and delivery typically takes place several weeks after the contract signing date. The transaction price is the dollar price of one $mtCO_2e$ at the USD exchange rate of the signing date, so transaction prices in our dataset are effectively near-term futures prices. The exact tenor within nine months is not a price factor, as Section 4 will show.

are not comparable to voluntary credits. Fourth, we drop about 100 transactions in credits from small local registries. The final sample contains about 7,200 transactions by around 1,200 unique customers across nearly 400 unique projects, covering about 7% of secondary VCM transactions by dollar value in the sample period.

3.2 Sample descriptives

Table 1 reports sample composition by year, customer industry, and project technology. Transaction counts grew between 2018 and 2021 and reverted to pre-2020 levels by the end of the sample, mirroring aggregate VCM trading dynamics (Figure 2). Industry shares are stable across years, apart from the manufacturing share in early years.[13] Around 40% of transactions are with buyers from heavy-polluting industries (energy, transportation, manufacturing); the unique-buyer share may differ if trade frequency varies by industry. Renewable energy projects account for 44% of the sample, followed by nature-based avoidance projects (22%) and household projects (14%); waste, nature-based removal, and industrial projects are less common, at 10%, 7%, and 3%, respectively.[14] Renewables and household shares are broadly stable across years; nature-based avoidance fluctuates between 13–31% (peaking around 2020) and nature-based removals grow from below 1% to roughly 10% over the sample.

Customers from different industries purchase broadly similar technology mixes (Table 2). Financial and consumer-sector firms buy relatively more nature-based removal credits, while manufacturing and transportation firms buy more renewables credits, but the differences are economically

[13]The industry classification is provided by the dealer and follows the taxonomy of Standard and Poor's (GICS). The 'consumer' sector here refers to the manufacturing and provision of consumer goods and services, whereas 'manufacturing' covers other industrial manufacturing activities, to avoid confusion with 'Industrial' emission-reduction technologies.

[14]The definitions of project technologies are not standardized. We follow the granular classification provided by the dealer, aggregated into broader categories following the Ecosystem Marketplace taxonomy. The 'Nature-based' category in the dealer's classification corresponds to 'Forestry and land use' in the Ecosystem Marketplace taxonomy. The 'industrial' projects category also contains transportation, chemical processes, and industrial carbon capture projects.

	2018	2019	2020	2021	2022	2023	2024	Total
No. transactions	556	795	940	1441	1361	1240	870	7203
(A) By customer industry, % of total								
Manufacturing	41.2	24.9	25.3	23.2	17.9	20.2	23.9	23.6
Consumer	15.3	23.4	30.3	19.4	23.2	19.9	17.5	21.5
Energy	5.0	6.7	7.2	10.8	7.2	9.7	6.7	8.1
Finance	11.3	18.0	14.0	22.1	26.7	21.0	27.0	21.0
Other	23.2	21.1	15.9	16.4	17.2	13.5	19.5	17.4
Transport	4.0	5.9	7.2	7.9	7.9	15.6	5.4	8.3
(B) By project technology, % of total								
Industrial	2.5	1.1	2.2	4.3	2.5	1.6	1.4	2.4
Renewables	49.8	41.5	39.1	45.5	47.6	42.1	44.6	44.3
NB Removals	0.9	3.1	6.9	6.5	7.6	10.0	9.9	7.0
NB Avoidance	15.8	26.4	30.9	24.6	21.8	19.2	13.1	22.1
Household	20.5	16.0	14.0	9.0	9.3	16.3	18.4	13.8
Waste	10.4	11.8	6.8	10.1	11.1	10.8	12.6	10.5

Table 1: **Transactions by year, customer industry, and mitigation technology.**
This table shows the sample composition over time, dividing transactions by the industry of the buyer (Panel A) and by the technology of the project (Panel B). Both customer industry and project technology classifications are in-house taxonomies of the data provider, yet the former closely follows the Standard and Poor's industry classification (GICS), while the latter is similar to the one by Ecosystem Marketplace. In particular, 'Industrial' projects are primarily aimed at recovering waste energy. 'Renewables' are mostly onshore wind turbines, solar, and hydro power plants. 'NB' stands for 'nature-based'. 'NB Removals' are afforestation, reforestation, and revegetation projects. 'NB Avoidance' is the protection of existing forests (REDD+). 'Household' primarily refers to projects aimed at improving cookstoves and household water filtering. 'Waste' refers to landfill gas, composting, industrial wastewater treatment, and manure management improvements.

	By project technology, % total						Total no.
	IND	RNW	NBR	NBA	HLD	WST	trades
Manufacturing	2.1	49.8	5.6	19.0	14.8	8.6	1702
Consumer	1.8	41.6	7.4	28.6	10.6	10.0	1551
Energy	4.0	46.3	3.3	25.1	9.6	11.7	581
Finance	3.5	37.1	10.2	21.8	17.7	9.8	1516
Other	1.8	44.4	8.0	20.3	13.0	12.4	1254
Transport	1.7	51.3	3.2	15.5	14.7	13.7	599
Total	2.4	44.3	7.0	22.1	13.8	10.5	7203

Table 2: **Transactions by customer industry and project mitigation technology.** This table shows the sample composition by project technology, separately for buyers grouped by industry. The project labels stand for the following: 'IND' – industrial projects, 'RNW' – renewables, 'NBR' – nature-based removals, 'NBA' – nature-based avoidance, 'HLD' – household, and 'WST' – waste. The numbers in each line sum up to the total number of transactions with customers from a given industry.

small. The mode of trades per buyer is 1, the median is 3, and the mean is 6.[15] The fraction of single-transaction buyers in the total number of unique customers varies from 5% for industrial projects to 24% for renewables.

About 60% of sample projects are located in Asia (predominantly renewable energy). African projects are mainly household interventions in least-developed countries. European projects make up only a small share, predominantly renewables. The Verra and Gold Standard registries account for 50% and 30% of the sample, respectively, with the remainder mostly from CDM, a legacy UN-mandated registry established in 2005 under the Kyoto Protocol.

Table 4 reports transaction-level summary statistics. The average price is \$7.4 per $mtCO_2e$—roughly 80–85% below the contemporaneous compliance prices in the EU ETS or California cap-and-trade system. Prices are highly variable: the 5th percentile is around \$1 per ton and the 95th close to \$20. The average transaction is for about 12 thousand $mtCO_2e$, equivalent to roughly \$90,000. The dealer classifies about a third of buyers as 'multinational companies' and another third as 'small and medium enterprises' (SMEs); about half of buyers are privately owned. Half of

[15] Customers may choose to retire purchased credits or resell them later. Retirement is executed by the dealer in approximately 79% of sample transactions, and this fraction is similar across industries (e.g., 80% for financial-sector buyers), suggesting that financial firms in our sample mostly purchase credits for redemption rather than for trading.

	IND	RNW	NBR	NBA	HLD	WST	Total
No. projects	14	194	36	30	77	45	396
(A) By project region							
Africa	1	4	8	8	40	0	61
Americas	1	20	14	17	5	4	61
Asia	10	149	14	5	32	38	248
Europe	2	21	0	0	0	3	26
(B) By tercile of country per capita GDP							
T1 (poorest)	3	67	12	10	61	4	157
T2	7	94	20	18	14	35	188
T3 (richest)	4	33	4	2	2	6	51
(C) By tercile of country carbon emissions							
T1 (green)	0	5	3	5	21	0	34
T2	1	17	12	11	29	2	72
T3 (brown)	13	172	21	14	27	43	290
(D) By project registry							
CDM	5	44	1	0	8	10	68
Gold Standard	2	52	3	0	61	7	125
Verra	7	98	32	30	8	28	203

Table 3: **Sample projects across project region and registry.**
This table shows the sample compositions of the decarbonization projects in the sample. Specifically, it shows the counts of unique VCM credit projects, separately for project region (Panel A) and project registry (Panel B). 'CDM' is the UN-mandated registry for 'certified emissions reductions'. Gold Standard and Verra are independent non-profit organizations. The project labels stand for the following: 'IND' – industrial projects, 'RNW' – renewables, 'NBR' – nature-based removals, 'NBA' – nature-based avoidance, 'HLD' – household, and 'WST' – waste.

transactions are settled within a month of signing (mean: two months); 5% are settled in non-major currencies (defined as those outside EUR, USD, GBP, AUD, or CHF), 2% are credits issued in the buyer's home country, and 10% of customers are explicitly committed to climate targets under the Science Based Targets initiative (SBTi).[16]

3.3 External validity

Although the dealer is large, our reliance on a single source raises a concern about external validity. We address this concern by drawing on an independent transaction dataset from Xpansiv CBL, an electronic trading platform, and S&P Platts, a commodity intelligence provider. Our core project-level pricing results replicate in this auxiliary dataset (Appendix Table A1), and—because

[16]The SBTi is a non-profit organization that provides guidance for setting corporate emission-reduction targets and validates them against pathways consistent with limiting global warming to 1.5°C above pre-industrial levels.

	Mean	S.D.	5th	25th	Med.	75th	95th	N.Obs.
Signing price, $ per mtCO2e	7.4	7.6	1.1	3.0	5.6	9.7	19.3	7203
Signing premium/discount to ETS, %	-84.9	23.7	-97.5	-94.7	-90.1	-82.9	-56.7	6912
Volume, thousand mtCO2e	12.2	39.5	0.0	0.4	1.8	8.0	52.7	7203
Delivery – signing date, days	53.7	69.6	1.0	8.0	21.0	68.0	220.0	7203
Delivery – vintage year, years	4.9	2.8	1.0	3.0	4.0	6.0	10.0	7203
Vintage – project launch, years	3.4	3.0	0.0	1.0	3.0	5.0	10.0	7203
Project size, thousand mtCO2e	323.5	810.0	0.0	0.0	59.5	264.0	1259.7	7203
Home bias (dummy)	0.02	0.15	0.00	0.00	0.00	0.00	0.00	7203
Customer commited to SBTi (dummy)	0.10	0.30	0.00	0.00	0.00	0.00	1.00	7203
Non-standard currency (dummy)	0.05	0.22	0.00	0.00	0.00	0.00	0.00	7203

Table 4: **Summary statistics for the sample of VCM transactions.**
This table shows summary statistics for the main variables in the sample. Signing price is the transaction price agreed between the seller (VCM dealer) and a buyer (customer), expressed in $ per $mtCO_2e$ at the exchange rate as of the signing date. Premium to ETS is the percentage difference between the transaction price and the price of the $mtCO_2e$ in the European Emission Trading System as of the signing date (negative values represent a discount: VCM prices are below ETS prices). Volume is the number of credits traded, in thousands. Delivery delay is the number of days from signing to delivery, truncated at 270. Vintage refers to the year in which the traded carbon credit was issued. Project launch is the start of the crediting period (earliest available vintage). Project size is the expected annual emission reduction at project launch (self-reported by project developers). 'Home bias' equals 1 when the carbon credit buyer is domiciled in the country of the emissions reduction project, and 0 otherwise. SBTi commitment is 1 if the customer is explicitly committed to SBTi targets, and 0 otherwise. 'Non-standard currency' is 1 for transactions not settled in EUR, USD, GBP, AUD, or CHF.

its transactions can be linked to public project identifiers—it also lets us extend the analysis with external project-level data not available for our anonymized core dataset.

The auxiliary sample (CBL+Platts hereafter) comprises approximately 3,900 single-name transactions between 2019 and 2024—roughly half the size of our core dataset. Of these, 57% are from CBL's standard limit-order-book (LOB) trading tape, 39% are pre-negotiated trades executed via CBL but outside the LOB, and 4% are from S&P Platts price assessments collected from market participants by industry analysts. We link CBL+Platts transactions to project-level data from AlliedOffsets, a comprehensive VCM project information database that aggregates project characteristics, credit-quality ratings from BeZero Carbon (available for a subset of transactions), and self-reported Sustainable Development Goal (SDG) contributions from project documentation.

CBL+Platts trades differ from our core sample on several dimensions: they are cheaper (\$5 vs. \$7.4 per mtCO_2e), larger (50.9 vs. 12.2 thousand mtCO_2e), and more concentrated in renewable energy and nature-based avoidance projects (85% vs. 66%; Table 5 and Appendix Table A2). Despite these differences, our key empirical findings replicate in the CBL+Platts sample: the technology-premium ranking, the volume discount, and the vintage discount all appear with comparable signs and magnitudes (Appendix Table A1, discussed in Section 4.2). The auxiliary sample also allows us to examine the relationship between prices and external quality and co-benefit measures, discussed in Section 4.2.[17]

[17] Liu (2025) obtains complementary results in a similar AlliedOffsets sample.

	Mean	S.D.	5th	25th	Med.	75th	95th	N.Obs.
Signing price, $ per mtCO2e	5.0	4.7	0.5	1.7	4.0	6.8	12.5	3,875
Volume, thousand mtCO2e	50.9	129.0	0.1	2.0	15.0	50.0	200.0	3,875
Delivery – vintage year, years	4.7	2.8	1.0	2.0	4.0	6.0	10.0	3,875
Vintage – project launch, years	4.3	3.1	0.0	2.0	4.0	7.0	10.0	3,875
Project size, thousand mtCO2e	1074.2	1681.3	23.9	95.9	264.1	1,348.7	5,671.6	3,875

Table 5: **Summary statistics for the CBL+Platts sample of VCM transactions.** This table is analogous to Table 4, but for the CBL+Platts sample. The signing price is the transaction price for single-name VCM credits on CBL or in Platts price assessments, expressed in USD per $mtCO_2e$ at the exchange rate as of the transaction date. Volume is the number of credits traded, in thousands. Vintage refers to the year in which the traded carbon credit was issued. Project launch marks the start of the crediting period (the earliest available vintage). Project size is the expected annual emission reduction at project launch (self-reported by project developers). The sample period is from 2019 to 2024.

4 Pricing in the voluntary carbon market

We document two pricing facts about the VCM: (i) large price dispersion across project types; and (ii) large price dispersion within a given project across buyers. We show that both project-level and buyer-level characteristics can explain a substantial part of the observed price variation. Taken together, these patterns are difficult to reconcile with the view that voluntary carbon credits trade as a homogeneous commodity.

4.1 Price dispersion

Transaction prices in the VCM vary by more than an order of magnitude at every point in time, ranging from a few cents to over $100 per $mtCO_2e$ (Figure 1 in the Introduction). The degree of VCM price variability, as measured by the coefficient of variation, exceeds 100%. To put this number into context, the average within-month coefficient of variation of property sales prices in the UK—a market with substantial heterogeneity in the underlying assets—is lower, at around 85%.[18]

[18]The underlying data are from the UK Land Registry Price Paid Data. To calculate the coefficient of variation, we truncate prices at the 0.5% and 99.5% percentiles per month, as in the VCM sample, considering transactions

VCM prices vary substantially between projects. Averaging prices by project, we find a price range from \$0.23 to \$36 per $mtCO_2e$ in 2024. For reference, at the same point in time, the wholesale price of wheat in the U.S. varied from \$6 to \$8 per bushel across wheat grades; the price of a barrel of crude oil varied from \$59 to \$72.[19]

The between-project price dispersion is systematically related to the project type. Renewable-energy and waste-management credits cluster at the low end of the price distribution, with averages around \$3–\$4 per $mtCO_2e$. Nature-based avoidance and household-level interventions trade at substantially higher levels, often above \$10 per $mtCO_2e$, with nature-based removals reaching the highest prices. Cross-project price differences of this magnitude are unusual for goods that purport to deliver a homogeneous output—one ton of CO_2e abatement.

Even within a given project and vintage, transaction prices differ substantially across buyers. Figure 3 illustrates this for the three projects with the largest number of transactions in our sample at credit vintages 2015–2017: two buyers may simultaneously pay prices that differ by a factor of two or more for credits issued by the same project in the same year. The dispersion is not driven by buyer industry alone: within a given industry, prices for the same project still differ substantially.

This pattern is systematic. Across the broader sample, three regularities stand out: (i) within-project dispersion increases with the number of distinct buyers transacting in a project, consistent with buyer-side heterogeneity driving the spread; (ii) the coefficient of variation is smaller for expensive technology categories such as nature-based and household projects than for cheaper technologies such as renewables, suggesting that dispersion is not mechanically driven by price levels; and (iii) within-project variation declined substantially after the introduction of standardized

between 2018 and 2024. Alternatively, taking the ratio of the 95th to the 5th price percentiles, we find a multiple of 20 for the VCM and 13 for the UK property market, again highlighting the larger dispersion in the VCM market.

[19] Data sources: U.S. Wheat Associates price report (January 2025) and Energy Transfer crude oil posted prices (January 2025).

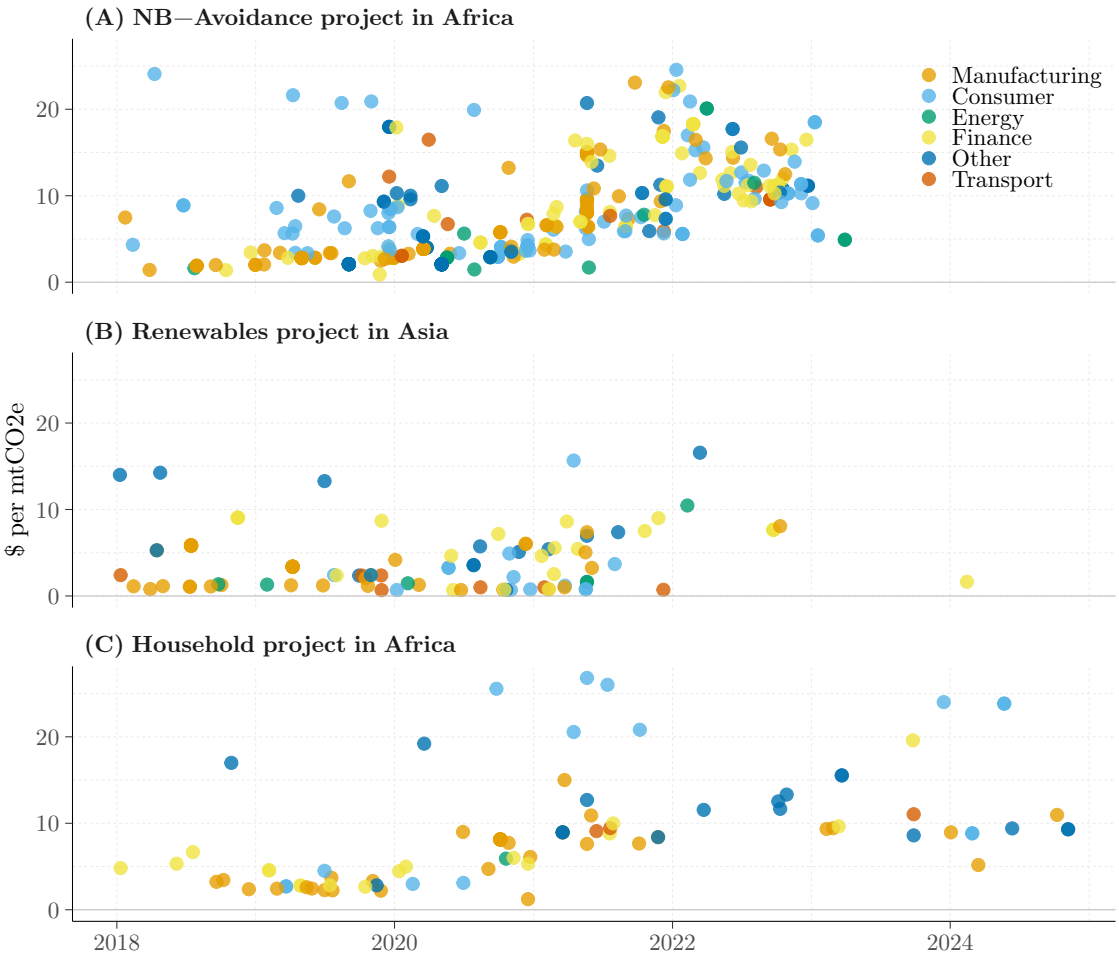


Figure 3: **VCM prices for individual projects (2015–2017 vintages), by client industry.** The dots are individual transaction prices from our core dealer-to-customer dataset, in $ per $mtCO_2e$. The three plotted projects are those with the largest number of transactions in 2015–2017 credit vintages (i.e., credits issued prior to the sample trading period) in the dataset. The dots are colored according to the customer's (credit buyer) industry affiliation.

contracts at Xpansiv CBL in the early 2020s, pointing to a role for pre-trade price transparency. We substantiate these results in Section 4.5 (Table 13).

4.2 Project-level pricing factors

If carbon credits are priced as differentiated goods (rather than as a homogeneous commodity), project-level characteristics should be reflected in transaction prices. We test this directly with a hedonic linear pricing model, shown in Equation (1):

$$\log(Price)_{i,p,t} = \alpha + \boldsymbol{\beta}\,\text{Project traits}_{p,t} + \boldsymbol{\gamma}\,\text{Transaction traits}_{i,p,t} + \delta_t + \Gamma_p + \Lambda_i + \epsilon_{i,p,t}. \quad (1)$$

The dependent variable, $\log(Price)_{i,p,t}$, is the natural logarithm of the price in \$/mtCO_2e at which customer i bought carbon credits from project p in year t. Project traits$_{p,t}$ is a vector of (time-varying) project characteristics that, in the strictest specification, include the project technology, its registry, issuing country GDP per capita and emissions, and the project age and size. Transaction traits$_{i,p,t}$ covers transaction characteristics, namely the log of the trading volume, the credit vintage, whether the transaction is denominated in a non-standard currency, delivery delay, and whether the credit was retired upon delivery. δ_t, Γ_p, and Λ_i are, respectively, year, project-country, and client (credit buyer) fixed effects. $\epsilon_{i,p,t}$ is the error term, clustered at the project level. Table 6 shows the results.

Project technology is the largest single project-level price factor. Five technology dummies alone explain 29% of price variation in the dealer-to-customer sample (Model 1, Table 6). Across the nine specifications, nature-based avoidance and household credits trade at premia of 92–134%

	Dependent variable: log (Price)								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Intercept	1.26***	1.25***	0.10	0.09	0.42				
	(0.05)	(0.08)	(0.38)	(0.37)	(0.35)				
Tech: Industrial	−0.15	−0.16	−0.17	−0.17	−0.10	−0.08	−0.09	0.03	−0.03
	(0.20)	(0.21)	(0.19)	(0.20)	(0.21)	(0.19)	(0.22)	(0.33)	(0.22)
Tech: NB Removal	1.43***	1.45***	1.34***	1.34***	1.34***	1.32***	1.48***	1.53***	1.21***
	(0.11)	(0.12)	(0.14)	(0.14)	(0.12)	(0.11)	(0.10)	(0.14)	(0.08)
Tech: NB Avoidance	0.78***	0.80***	0.76***	0.76***	0.69***	0.74***	0.83***	0.75***	0.78***
	(0.08)	(0.10)	(0.10)	(0.10)	(0.09)	(0.07)	(0.12)	(0.18)	(0.11)
Tech: Household	0.84***	0.81***	0.84***	0.84***	0.77***	0.71***	0.76***	0.85***	0.65***
	(0.08)	(0.09)	(0.10)	(0.10)	(0.09)	(0.07)	(0.08)	(0.11)	(0.07)
Tech: Waste	0.08	0.09	0.05	0.05	0.06	0.08	0.14*	0.17*	0.004
	(0.11)	(0.11)	(0.09)	(0.09)	(0.09)	(0.09)	(0.08)	(0.09)	(0.07)
Registry: Gold Standard		0.05	0.02	0.02	−0.01	0.10	0.31***	0.30***	0.12*
		(0.09)	(0.09)	(0.09)	(0.08)	(0.08)	(0.08)	(0.10)	(0.07)
Registry: Verra		−0.02	−0.03	−0.04	−0.01	0.02	0.05	0.09	−0.07
		(0.10)	(0.09)	(0.10)	(0.09)	(0.08)	(0.08)	(0.11)	(0.06)
log (GDP p.c.)			0.18***	0.18***	0.17***	0.11**	−0.42	−0.71	0.03
			(0.05)	(0.05)	(0.05)	(0.05)	(0.59)	(0.72)	(0.50)
log (Country emissions)			−0.06***	−0.06***	−0.06***	−0.05***	−0.08	−0.20	−0.45
			(0.02)	(0.02)	(0.02)	(0.02)	(0.38)	(0.60)	(0.33)
Project age (years)				0.002	0.01	0.004	0.005	0.001	−0.001
				(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
Project size (std. per tech)				0.002	0.002	−0.02	−0.05**	−0.07**	−0.03*
				(0.04)	(0.03)	(0.03)	(0.02)	(0.03)	(0.02)
log (Trade volume)					−0.09***	−0.10***	−0.09***	−0.08***	−0.05***
					(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
Credit vintage (years)					−0.02*	−0.03***	−0.02***	−0.02**	−0.02***
					(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
Non-standard currency (dummy)					0.23***	0.35***	0.36***	0.30***	0.12
					(0.05)	(0.07)	(0.07)	(0.11)	(0.10)
Delivery delay (months)					−0.01	−0.01	−0.005	0.002	0.01
					(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
Non-retirement delivery (dummy)					−0.33***	−0.25***	−0.23***	−0.22***	0.01
					(0.04)	(0.03)	(0.03)	(0.04)	(0.05)
Year FEs	NO	NO	NO	NO	NO	YES	YES	YES	YES
Country FEs	NO	NO	NO	NO	NO	NO	YES	YES	YES
Clients with 10+ trades	NO	NO	NO	NO	NO	NO	NO	YES	YES
Client FEs	NO	NO	NO	NO	NO	NO	NO	NO	YES
Observations	7,203	7,203	7,203	7,203	7,203	7,203	7,203	3,974	3,974
Adjusted R^2	0.29	0.29	0.32	0.32	0.42	0.52	0.59	0.60	0.79

Table 6: **Panel models for VCM transaction prices and project characteristics.** This table shows OLS regression estimates that explore which project and transaction factors explain VCM transaction prices. The dependent variable is the natural logarithm of the transaction price. Technology and registry indicators correspond to the categories in Table 3. The reference (dropped) categories are, respectively, renewables projects (tech) and CDM (registry). Annual GDP per capita (expressed in constant 2021 USD) and country emissions are from the World Bank WDI dataset. Project age is the difference in years between the traded certificate's vintage year and the project launch (the earliest available vintage per project). Project size is the total emission reduction expected at project launch, standardized to zero mean and unit variance per project technology. The carbon credit vintage is the year difference between the transaction date and the credit's vintage year. The non-standard currency indicator identifies transactions settled in a currency other than EUR, USD, GBP, AUD, or CHF. Delivery delay is the difference in months between the signing and delivery dates. 'Non-retirement delivery (dummy)' takes the value of 1 if the credit was transferred to the buyer without immediate retirement by the dealer on behalf of the client. Models 8 and 9 are estimated using the subsample of customers with at least 10 transactions. The standard errors (in parentheses) are clustered at the project level; $^{*}p$ <0.1; $^{**}p$ <0.05; $^{***}p$ <0.01.

relative to renewables, and nature-based removals at 235–362%.[20] The same ranking and similar magnitudes appear in the CBL+Platts sample: 67–186% for nature-based avoidance and 148–297% for nature-based removals (Appendix Table A1). The largest premia accrue to project types whose climate value is widely viewed as weakest. We return to this misalignment in the next subsection. Remarkably, technology premia remain large and significant in the subset of trades with repeated credit buyers and after controlling for client fixed effects (Models 8 and 9, respectively), suggesting that buyer selection into preferred technologies does not drive the premia.[21]

Among the three registries in our sample, only Gold Standard certification commands a price premium, and only after controlling for the project host country. Model 2 adds registry dummies for Gold Standard and Verra credits, both relative to CDM. The Gold Standard premium is insignificant in the pooled specification but is statistically significant once country fixed effects are added, reaching 13–36% across Models 7 to 9. Verra credits carry no significant premium over CDM credits in any specification.

Project prices reflect host-country emissions and income. In Model 3, a 10% increase in host-country emissions is associated with a 0.6% lower price, and a 10% increase in per capita GDP with a 1.8% higher price. We read the latter as an installation-cost effect within technology. The host-country emissions and per capita GDP effects are absorbed by country fixed effects (Models 7–9), consistent with a level effect across countries rather than a within-country panel relation.

Project age has no pricing effect, while larger projects trade at lower prices, controlling for project host country, consistent with a supply-side cost effect within technology. The age coefficient—years between project launch and credit issuance—is always statistically insignificant. Project size, standardized within technology, is near zero and insignificant in the pooled and year-FE

[20] Price premia and discounts reported hereafter are after log-adjustment, i.e., a 92% household premium is $100\% \times (e^{0.65} - 1)$ in Model 9 of Table 6.

[21] We elaborate on buyer-related credit price characteristics in Section 4.3.

specifications (Models 4–6) but turns negative and significant once country fixed effects are added: a one-standard-deviation increase relates to 3–7% lower prices (Models 7–9).

Transaction-level pricing in the VCM mirrors patterns familiar from other OTC markets. A 10% increase in trade volume is associated with a 0.8–1% lower transaction price (Models 5–8), consistent with dealer inventory management and client-relationship pricing (e.g., Pinter et al., 2024). Holding client identity constant, the within-client elasticity falls to 0.5% (Model 9), suggesting that inventory and client-relationship management effects contribute equally to the volume discount. Transactions settled in currencies outside the major foreign-exchange pairs trade at a 26–43% premium, consistent with the dealer pricing in the cost of hedging FX risk. Model 10 shows that this premium is mostly due to a few clients who settle exclusively in non-standard currencies: the FX premium is statistically insignificant and smaller in magnitude, controlling for customer identity.

VCM transaction prices also reflect transaction-level features specific to carbon credits, such as credit vintage and retirement status. Older vintages, measured as the years between the transaction date and credit issuance, carry a 2–3% discount per additional year across Models 5–9, consistent with concerns that credits issued under earlier, less stringent methodologies are less reliable. Credit transfers without immediate retirement, about 21% of sample transactions, trade 25–39% cheaper than the remaining 79% in which the dealer retires the credit on the client's behalf, possibly reflecting speculative rather than offsetting use.[22] The non-retirement discount, like the FX premium, disappears in the model specification with client fixed effects (Model 10), suggesting that it is also due to a few clients who prefer a particular type of settlement. Delivery delay carries no significant effect. Together, the transaction-level variables add 10% to the explained price variation in addition to project characteristics—roughly a third of what project technology alone explains.

[22] Appendix Table A3 shows that the price premia and discounts associated with project technologies remain largely unchanged when estimated in the subset of transactions without credit retirement.

With year and country fixed effects, the hedonic specification explains 59% of VCM price variation (60% in the subset of actively-trading customers), a substantial share for an OTC market without centralized price discovery. Client-level factors add almost 20% to the model's explanatory power (Model 10 compared to Model 9) — a remarkable result, which we put into a broader context in Section 4.3. Yet the large technology premia documented above persist across all specifications, raising the question of what they represent.

Technology premia and credit quality

The technology premia documented above are not aligned with the expected climate value of the underlying credits. Table 7 maps the premia from Table 6 onto three external quality benchmarks: the delivery-to-promise ratio in Probst et al. (2024), which assesses the share of credited emission reductions that materialize; the additionality assessment in Cames et al. (2017), which captures whether projects would likely have proceeded without carbon-credit financing; and whether the project type has an approved methodology under the UNFCCC Article 6.4 mechanism of the Paris Agreement, the UN's most recent carbon crediting mechanism with a focus on high-quality projects.

Premia and external reliability assessments are misaligned across technology categories. Nature-based avoidance and household credits command the largest premia in the rated set, even though both score poorly on delivery-to-promise, and household credits also score poorly on additionality. Industrial credits trade at no statistically significant premium despite higher delivery-to-promise and stronger additionality. Waste credits trade at no significant premium either, although they are the only project type in the table with an approved Article 6.4 methodology and receive a more favorable additionality assessment compared to renewable credits. Nature-based removals carry the largest premium in our data but are not covered by either external benchmark, likely because such projects have only recently gained prominence.

Technology	Price premium (vs. Renewables)	Delivery-to-promise (Probst et al., 2024)	Additionality (Cames et al., 2017)	Article 6.4 Methodology
NB Removals	+274–362%	—	—	—
NB Avoidance	+99–146%	0–24%	—	—
Household	+103–159%	11%	Low	—
Industrial	Insignificant	16–68%	Medium-high	—
Waste	Insignificant	—	Medium	✓
Renewables	n.a. (Reference technology)	0%	Low	—

Table 7: **Technology premia and carbon credit quality.**
This table compares estimated technology price premia from Table 6 with external assessments of carbon credit quality. Delivery-to-promise denotes the share of credited emission reductions that materialize, as estimated by Probst et al. (2024). Additionality reflects the likelihood that the project would not have been undertaken without carbon-credit financing, as assessed by Cames et al. (2017). Article 6.4 Methodology indicates whether the project type has an approved methodology under the UNFCCC Article 6.4 mechanism of the Paris Agreement.

A direct project-level quality measure delivers a similar conclusion. For a subset of transactions in the CBL+Platts sample, we observe project credit ratings from the rating agency BeZero, which evaluates the likelihood that the promised emission reductions will materialize.[23] In Table 8, we add BeZero ratings on a numeric scale from 1 (AA) to 6 (below B) to the baseline model (1). Ratings are unconditionally related to prices: a project rated one notch lower trades 22% lower on average (Model 2). They also absorb part of the nature-based avoidance premium, which falls from 92% to 37% (still significant). Once project-country and transaction-year fixed effects are added (Models 6–7), however, BeZero ratings lose statistical significance, and the nature-based premia recover to 51–98%. Project ratings appear to reflect average differences across host countries rather than systematic quality differences across project technologies.

[23] Project additionality is part of creditworthiness in the BeZero methodology; project co-benefits are not. Out of 602 CBL+Platts transactions with available BeZero ratings, about 45% are in AA-, A-, or BBB-rated credits, with the remainder rated BB or lower.

	Dependent variable: log (Price)						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Intercept	1.96	3.07***	1.39	2.55**			
	(1.29)	(1.14)	(1.21)	(1.05)			
Tech: Industrial	0.11	−0.28*	0.04	−0.27*	−0.21	−0.28	−0.34
	(0.21)	(0.15)	(0.18)	(0.15)	(0.13)	(0.32)	(0.26)
Tech: NB Removal	1.27***	0.92***	1.17***	0.91***	1.01***	0.98***	0.97***
	(0.33)	(0.27)	(0.30)	(0.27)	(0.32)	(0.18)	(0.19)
Tech: NB Avoidance	0.92***	0.37**	0.78***	0.36**	0.50***	0.54**	0.51**
	(0.20)	(0.17)	(0.20)	(0.17)	(0.16)	(0.22)	(0.22)
Tech: Household	0.69**	0.47**	0.49*	0.38*	0.80***	0.04	0.24
	(0.27)	(0.20)	(0.28)	(0.21)	(0.17)	(0.31)	(0.26)
Tech: Waste	1.51***	1.73***	1.27**	1.55***	1.65***		
	(0.56)	(0.43)	(0.52)	(0.42)	(0.43)	(0.00)	(0.00)
Registry: Gold Standard	−0.07	−0.30	0.11	−0.15	−0.45*	1.15***	0.92***
	(0.36)	(0.32)	(0.38)	(0.32)	(0.25)	(0.33)	(0.31)
Registry: Verra	−0.54	−0.64**	−0.38	−0.53*	−0.84***	0.67***	0.53**
	(0.35)	(0.32)	(0.35)	(0.31)	(0.26)	(0.25)	(0.21)
log (GDP p.c.)	0.03	0.10	0.10	0.13	0.12	−11.04***	−2.39
	(0.15)	(0.12)	(0.12)	(0.11)	(0.11)	(2.55)	(2.69)
log (Country emissions)	−0.03	−0.08***	−0.06	−0.09***	−0.05*	−0.85	4.41
	(0.06)	(0.03)	(0.04)	(0.03)	(0.03)	(3.71)	(2.96)
Project age (years)	−0.01	−0.02	−0.01	−0.02	0.003	−0.03	−0.02
	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)
Project size (std. per tech)	0.06	0.004	0.08	0.02	0.05	−0.03	−0.04
	(0.07)	(0.05)	(0.08)	(0.06)	(0.05)	(0.04)	(0.03)
log (Trade volume)	−0.005	0.004	0.001	0.01	−0.04***	−0.02*	−0.04***
	(0.02)	(0.02)	(0.02)	(0.02)	(0.01)	(0.01)	(0.01)
Credit vintage (years)	−0.12***	−0.13***	−0.13***	−0.13***	−0.10***	−0.10***	−0.10***
	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)
BeZero project rating (1–6)		−0.22***		−0.19***	−0.16***	−0.04	−0.03
		(0.05)		(0.04)	(0.04)	(0.05)	(0.05)
No. SDGs in the PDD			0.04**	0.02***	0.02**	0.001	0.001
			(0.02)	(0.01)	(0.01)	(0.01)	(0.005)
Year FEs	NO	NO	NO	NO	YES	NO	YES
Country FEs	NO	NO	NO	NO	NO	YES	YES
Observations	602	602	602	602	602	602	602
Adjusted R^2	0.28	0.36	0.32	0.38	0.57	0.59	0.64

Table 8: **Regression models for transaction log-prices in the CBL+Platts sample (only projects with BeZero ratings).**
This table is analogous to Table 6, but for the CBL+Platts sample where BeZero project ratings are available. The dependent variable is the natural logarithm of the transaction price. The definitions of explanatory variables are as in the main analysis in Table 6, except for the new variables: BeZero rating and the count of UN Sustainable Development Goals (SDGs) in the project development documentation (PDD). BeZero ratings are on the numerical scale: 1 (AA), 2 (A), 3 (BBB), 4 (BB), 5 (B), and 6 (lower than B). Higher BeZero scores indicate lower credit quality. 'No. SDGs in the PDD', a proxy for project co-benefits, is the variable that counts the number of UN SDGs mentioned in the PDDs, excluding SDG 13 ('Climate Action' – present in every VCM project PDD). The standard errors (in parentheses) are clustered at the project level; $^{*}p<0.1$; $^{**}p<0.05$; $^{***}p<0.01$.

Quality signals, therefore, enter VCM prices in some specifications—registry premia in Table 6 and the unconditional BeZero coefficient are clear examples—but the technology premia themselves are not ordered by the expected reliability of the promised emission reductions.

A potential concern is that the external assessments in Table 7 may not have been available, or salient, when many sample transactions occurred. If technology premia reflect the imperfect quality of information, they should change when new public information alters beliefs about delivery risk. Two pieces of evidence speak against this learning channel.

First, we exploit a salient negative information event for nature-based avoidance credits. On October 16, 2023, *The New Yorker* published an article documenting serious quality concerns about a large-scale nature-based avoidance project in Zimbabwe.[24] The article received broad attention in financial and policy media and saliently raised quality concerns about nature-based avoidance projects. Table 9 estimates whether the nature-based avoidance premium, relative to all other technologies, changed after that date. Across specifications, the interaction between nature-based avoidance and the post-event period is statistically indistinguishable from zero. These results are robust to focusing on an earlier article documenting similar shortcomings for a wide set of nature-based avoidance projects, published in *The Guardian* on January 18, 2023 (see Table A4). The premium for nature-based avoidance credits does not decline following a prominent public signal of delivery risk.

Second, the price premia appear to be stable across the full sample period in Appendix Figure A1, which reports the technology premia from a dynamic state-space model of VCM log-prices in the spirit of Francke and Vos (2004). The public information environment shifted substantially over the sample, including academic evidence on limited delivery-to-promise and several high-profile critiques of nature-based projects, yet the estimated technology premia are essentially flat.

[24] See The Great Cash-for-Carbon Hustle, October 16, 2023.

	Dep. var.: log (Price)		
	(1)	(2)	(3)
NB Avoidance (dummy)	0.50***	0.53***	0.37***
	(0.09)	(0.09)	(0.08)
Post NY (dummy)		0.004	−0.12*
		(0.09)	(0.07)
NB Avoidance × Post NY		−0.22	−0.01
		(0.22)	(0.10)
Year FE	YES	YES	YES
Client FE	NO	NO	YES
Observations	7,203	7,203	7,203
Adjusted R^2	0.32	0.32	0.68

Table 9: **Nature-based avoidance price premia around a public quality controversy.** This table estimates how the price premium of nature-based avoidance projects, relative to all other projects in the dealer-to-customer sample, changes after the October 16, 2023 publication of a *New Yorker* article documenting quality concerns about a large-scale nature-based avoidance project in Zimbabwe. 'Post NY' equals one after October 16, 2023 and zero otherwise. All models include calendar-year fixed effects. There are 81 transactions in the 'NB Avoidance×Post NY' group in 2023. Control variables are registry dummies, transaction volume, vintage, non-standard delivery currency, and non-retirement delivery. The standard errors (in parentheses) are clustered at the project level; $^{*}p$ <0.1; $^{**}p$ <0.05; $^{***}p$ <0.01.

Together, the cross-sectional and time-series evidence is difficult to reconcile with delivery risk as the primary driver of cross-project price dispersion. We next consider co-benefits as a potential explanation.

Technology premia and co-benefits

If credit quality does not explain away the technology premia, buyers may be paying for non-carbon project attributes. Co-benefits, i.e., social or environmental positive externalities arising from carbon projects, are the most prominent example. Improvements in household cooking technologies in low-income countries can raise local quality of life beyond emission reductions, and biodiversity gains from afforestation extend beyond carbon sequestration. VCM credit buyers may value such positive externalities and pay a premium for projects that deliver them.

Co-benefits are difficult to measure. They are typically reported by project developers but rarely evaluated by third parties. We proxy for them with the number of distinct UN Sustainable Development Goals (SDGs) that project developers claim to contribute to in the project development documentation. We exclude SDG 13 ('Climate Action') from the count since it is claimed in every project considered; other SDGs are mentioned in between 19% (SDG 14: Life Below Water) and 75% (SDG 8: Decent Work) of traded projects, with an average of 2.4 different SDGs per project. Returning to the CBL+Platts sample of Table 8, an extra SDG mentioned in the documentation is associated with a 4% price premium (Model 3). The SDG count moderately reduces the technology premium, but its effect largely disappears once country and transaction-year fixed effects are added (Models 6–7).

Co-benefits, as we measure them, enter VCM prices but do not absorb the technology premia. The SDG count is a coarse proxy: it records claims rather than delivered benefits, and weights goals equally regardless of how strongly buyers value each one. Technology categories likely capture additional dimensions of buyer value (e.g., salience, narrative appeal, and reputational signal) that the SDG count does not fully reflect.

4.3 Customer-level pricing factors

We next ask whether buyer characteristics are reflected in VCM transaction prices, conditional on the credits being purchased. In OTC markets, dealers may exercise market power and price discriminate across customers when buyers have limited ability to compare offers (e.g., Graddy, 1995; Li and Schürhoff, 2019). In that setting, buyer characteristics can affect prices even after holding project and transaction attributes fixed.

To evaluate price premia associated with different customer characteristics, net of project characteristics, we estimate the OLS model in Equation (2):

$$\log(Price)_{i,p,t} = \boldsymbol{\beta}\,\text{Client traits}_{i,t} + \boldsymbol{\psi}_{i,p,t} + \delta_t + \epsilon_{i,p,t}. \tag{2}$$

The main explanatory variables of interest are captured in Client traits$_{i,t}$, a vector of buyer's i characteristics, that include buyer industry, size, commitment to the SBTi, buyer's home country GDP per capita and emissions, and an indicator for projects from the client's home country. $\boldsymbol{\psi}_{i,p,t}$ is the set of project and trade characteristics from Equation (1), including project fixed effects in some specifications. δ_t captures year fixed effects, and $\epsilon_{i,p,t}$ is the error term, clustered at the project level. The results are presented in Table 10. Models 1–7 are estimated on the full sample, whereas Models 8 and 9 are estimated on the subset of returning clients with at least 10 transactions in the dataset.

Top-tier clients receive large price discounts, and the buyer industry is also strongly related to VCM prices. Table 10 presents the results. Top-tier clients, defined as the 20 largest accounts by trading volume, pay 16–23% less than other buyers beyond the transaction-level volume discount. Financial firms pay 9–17% more than industrial manufacturing firms, and consumer-sector firms pay 13–22% more, net of project-level factors. These premia remain economically large in the repeated-buyer subsample, suggesting that they are not driven by clients that appear only once or twice in the data. SMEs pay about 6% less than large firms in the full sample, although the SME discount is not statistically significant in the repeated-buyer subsample.[25]

Prices are also higher for buyers from wealthier countries and for credits generated in the buyer's home country. Clients from countries with 10% higher GDP per capita pay 0.9–1.3% more for the same credits in Models 4–7. The coefficient remains positive in the repeated-buyer subsample, but becomes statistically insignificant; once client fixed effects are included in Model 9, it flips sign.

[25] SMEs still comprise about one-third of unique buyers in the repeated-buyer subsample.

This pattern is consistent with dealers charging higher prices to clients from wealthier markets, rather than the same client paying more when its home-country income changes. We also find a home-country premium: credits whose project country matches the buyer's domicile trade at a 32–33% premium. This premium becomes statistically insignificant once client fixed effects are included, indicating that it reflects differences between buyers who purchase home-country credits and those who do not, rather than a within-buyer premium for local projects. This interpretation is consistent with Pedraza et al. (2026), who document that firms retire offsets generated in the countries where they operate, as they value the positive externalities associated with offsetting.

Explicit climate commitments do not predict higher prices. We proxy such commitments with SBTi commitment, the dominant framework for third-party validation of corporate emission reduction targets. The SBTi coefficient is small and statistically insignificant in every specification in Table 10. This null is less surprising in light of SBTi rules: carbon credits cannot be counted as emission reductions toward progress on near- or long-term science-based targets; they can only be used to neutralize residual emissions or to finance mitigation beyond a firm's science-based emission reduction targets.[26] Firms with an SBTi target may nevertheless purchase credits to substantiate environmental claims that go beyond their SBTi commitments, e.g., product-level carbon neutrality. Buyer-country emissions show a small negative association with prices in the full sample, but the relation is not robust in the repeated-buyer subsample or with client fixed effects. As discussed in Section 4.2, volume and vintage discounts are largely robust to controlling for buyer identity, whereas settlement currency and non-retirement credit delivery are not (Model 9).

Client identity accounts for a large share of buyer-side price variation. Comparing Models 8 and 9 in Table 10, the adjusted R^2 in the repeated-buyer subsample rises from 0.72 to 0.84 once client fixed effects are included. By contrast, in the full sample, observable client characteristics—size,

[26] See the SBTi Corporate Net-Zero Standard V1.3.1, criterion C12.

	Dependent variable: log (Price)								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
log (Trade volume)	−0.08***	−0.09***	−0.09***	−0.09***	−0.09***	−0.09***	−0.07***	−0.06***	−0.04***
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.005)
Credit vintage (years)	−0.02***	−0.02***	−0.02***	−0.02***	−0.02***	−0.02***	−0.02***	−0.02	−0.01**
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
Non-standard currency (dummy)	0.35***	0.36***	0.36***	0.37***	0.35***	0.34***	0.39***	0.36***	0.08
	(0.07)	(0.07)	(0.07)	(0.07)	(0.07)	(0.07)	(0.06)	(0.10)	(0.09)
Non-retirement delivery (dummy)	−0.13***	−0.12***	−0.12***	−0.11***	−0.10***	−0.10***	−0.13***	−0.10**	−0.01
	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.05)	(0.06)
Industry: Consumer	0.19***	0.17***	0.17***	0.16***	0.16***	0.15***	0.12***	0.20***	
	(0.04)	(0.04)	(0.04)	(0.04)	(0.04)	(0.04)	(0.03)	(0.05)	
Industry: Energy	−0.07	−0.10*	−0.09*	−0.08	−0.09	−0.10*	−0.11**	−0.01	
	(0.06)	(0.06)	(0.06)	(0.06)	(0.06)	(0.06)	(0.05)	(0.07)	
Industry: Finance	0.16***	0.15***	0.15***	0.14***	0.14***	0.13***	0.09***	0.12*	
	(0.04)	(0.04)	(0.04)	(0.04)	(0.04)	(0.04)	(0.03)	(0.07)	
Industry: Other	0.16***	0.14***	0.14***	0.13***	0.13***	0.12***	0.10***	0.11**	
	(0.04)	(0.04)	(0.04)	(0.04)	(0.04)	(0.04)	(0.03)	(0.05)	
Industry: Transport	0.04	0.02	0.03	0.03	0.01	0.01	−0.02	0.04	
	(0.05)	(0.05)	(0.05)	(0.05)	(0.05)	(0.05)	(0.05)	(0.08)	
Top-tier client (dummy)	−0.21***	−0.20***	−0.20***	−0.20***	−0.19***	−0.19***	−0.16***	−0.15***	
	(0.05)	(0.05)	(0.05)	(0.05)	(0.05)	(0.05)	(0.05)	(0.05)	
Size: Multinational		−0.01	−0.01	−0.01	−0.01	−0.004	−0.02	−0.03	
		(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.04)	
Size: SME		−0.06**	−0.06**	−0.06**	−0.06*	−0.06**	−0.06*	−0.07	
		(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.05)	
SBTi commitment			0.04	0.04	0.03	0.04	0.02	−0.03	
			(0.04)	(0.04)	(0.04)	(0.04)	(0.04)	(0.06)	
log (Country GDP p.c.)				0.10***	0.09***	0.13***	0.12***	0.11	−0.36
				(0.03)	(0.03)	(0.03)	(0.03)	(0.07)	(0.81)
log (Country emissions)					−0.02**	−0.02***	−0.02**	−0.01	−0.19
					(0.01)	(0.01)	(0.01)	(0.01)	(0.45)
Home bias						0.36***	0.28**	0.29	0.09
						(0.12)	(0.13)	(0.32)	(0.19)
Year FEs	YES	YES	YES	YES	YES	YES	YES	YES	YES
Project-level controls	YES	YES	YES	YES	YES	YES	NO	NO	NO
Project FEs	NO	NO	NO	NO	NO	NO	YES	YES	YES
Clients with 10+ trades	NO	NO	NO	NO	NO	NO	NO	YES	YES
Client FEs	NO	NO	NO	NO	NO	NO	NO	NO	YES
Observations	7,203	7,203	7,203	7,203	7,203	7,203	7,203	3,974	3,974
Adjusted R^2	0.60	0.60	0.60	0.60	0.60	0.61	0.68	0.72	0.84

Table 10: **Panel models for VCM transaction prices and buyer characteristics.**
This table shows OLS regression estimates that explore which client characteristics explain VCM transaction prices. Models 1–6 control for project size, mitigation technology, domicile country, registry, and transaction year fixed effects. Models 7–9 replace project-level controls with project fixed effects. Volume, carbon credit vintage, currency, delivery type, and project size variables are the same as those in the models in Table 6. 'Top-tier client' takes the value of 1 if the customer is in the top-20 trading accounts by transaction volume in the entire dataset. The reference (dropped) category for the buyer industry is industrial manufacturing, while for buyer size, it is large firms. GDP per capita and emissions are for the *buyer* domicile country (as per billing address). Models 8 and 9 are estimated using the subsample of customers with at least 10 transactions. The standard errors (in parentheses) are clustered at the project level; $^{*}p$ <0.1; $^{**}p$ <0.05; $^{***}p$ <0.01.

SBTi commitment, home-country GDP per capita, country emissions, and home bias—raise the adjusted R^2 by only about one percentage point (from 0.60 in Model 1 to 0.61 in Model 6). Thus, observable customer characteristics explain only part of the price variation we document, consistent with dealers also drawing on finer or relationship-specific information.

4.4 Comparing project- and client-level factors

The two preceding subsections show that project-level and client-level characteristics each explain economically large and partly overlapping shares of VCM price variation. Table 11 decomposes the explained variance into project-, transaction-, and buyer-related components, sequentially adding factor groups in different orders to bound the marginal contribution of each.

Project technologies explain a sizable share of VCM price variation, but buyer-related factors—especially buyer fixed effects—explain relatively more. Table 11 demonstrates this by sequentially adding project-, transaction-, and customer-level factors to the price Equations (1) and (2) in different orders. The explanatory power of project technology is bounded from above by 29% (marginal adjusted R^2 in Model 2: the regression of prices on technology dummies only), but the lower bound is at most 11% (Model 13: project technology is added on top of buyer characteristics, buyer fixed effects, all transaction-level controls, but before other project characteristics). In contrast, buyer-related factors are large by every cut: buyer IDs add 17 percentage points on top of all other factors (Model 8 minus Model 7); buyer characteristics and buyer IDs jointly add 19 percentage points on top of project and transaction factors (Model 8 minus Model 6); and a regression of prices on buyer characteristics and buyer IDs alone reaches 62% adjusted R^2 (Model 10).

Even by OTC-market standards, buyer fixed effects explain an unusually large share of variation in VCM prices. To benchmark this, Figure A2 presents the results of a similar analysis on a same-sized sample of U.S. corporate bond transactions with known buyer identities. In that sample,

	RSS	RSS (%)	RSE	RSE (%)	$Adj.R^2$	$\Delta Adj.R^2$	DF
(M1): Constant	5426	100.0	0.868	100.0	0.000		7202
(A) From project techonolgy to buyer ID							
(M2): M1+Project tech.	3833	70.6	0.730	84.1	0.293	0.293	7197
(M3): M2+Proj. country char.	3234	59.6	0.673	77.6	0.398	0.105	7133
(M4): M3+Proj. ID	2429	44.8	0.597	68.8	0.526	0.128	6805
(M5): M4+Transact. year	2047	37.7	0.549	63.2	0.600	0.074	6799
(M6): M5+Other trans. char.	1724	31.8	0.504	58.0	0.663	0.063	6794
(M7): M6+Buyer char.	1661	30.6	0.495	57.0	0.675	0.012	6783
(M8): M7+Buyer ID	639	11.8	0.338	39.0	0.848	0.173	5584
(B) From buyer charachteristics to project technology							
(M9): M1+Buyer char.	4868	89.7	0.823	94.8	0.101	0.101	7191
(M10): M9+Buyer ID.	1719	31.7	0.536	61.8	0.618	0.517	5974
(M11): M10+Transact. year	1647	30.4	0.525	60.5	0.634	0.016	5968
(M12): M11+Other trans. char.	1535	28.3	0.507	58.5	0.658	0.025	5963
(M13): M12+Proj. tech.	1038	19.1	0.417	48.1	0.769	0.111	5958
(M14): M13+Proj. country char.	845	15.6	0.379	43.6	0.810	0.041	5896
(M15): M14+Proj. ID	639	11.8	0.338	39.0	0.848	0.038	5584

Table 11: **VCM log-price variance decomposition.**
Each line is an OLS model of VCM transaction log-prices on specified variables. Model 1 (M1) is a regression on a constant. In part (A), we sequentially add (to Model 1) project, transaction, and buyer characteristics. In part (B), we proceed in reverse order. Other project characteristics are registry dummies, project size, and project age. Project country characteristics are country dummies, project country GDP, and project country CO_2 emissions. The transaction year is a set of calendar year dummies. Transaction characteristics include volume, credit vintage, a non-standard-currency dummy, and delivery delay. Buyer characteristics are industry affiliation (dummy for each industry), size, SBTi commitment, top-tier-client dummy, and the buyer's country GDP and emissions. RSS is the sum of squared model residuals. RSE is the square root of the ratio of RSS to model degrees of freedom (DF). RSS (%) and RSE (%) are relative to Model 1. $\Delta Adj.R^2$ is the change in $Adj.R^2$ (Adjusted R^2) relative to a preceding model.

despite a higher number of unique customers, buyer fixed effects explain only 15% of variation in bond prices when added first, and they add no explanatory power at all when added on top of known price factors. The pronounced role of buyer-related factors in explaining VCM price variation is consistent with dealers price-discriminating across buyers with heterogeneous willingness to pay. The market's limited pre- and post-trade transparency may be conducive to such discrimination.

4.5 An attempted commoditization

The VCM trades not only credits from individual projects ('single-name' credits) over-the-counter but also 'basket products' on exchanges. The most prominent spot VCM basket products are Xpansiv CBL's GEO contract family: GEO, N-GEO, and C-GEO contracts, introduced in 2020,

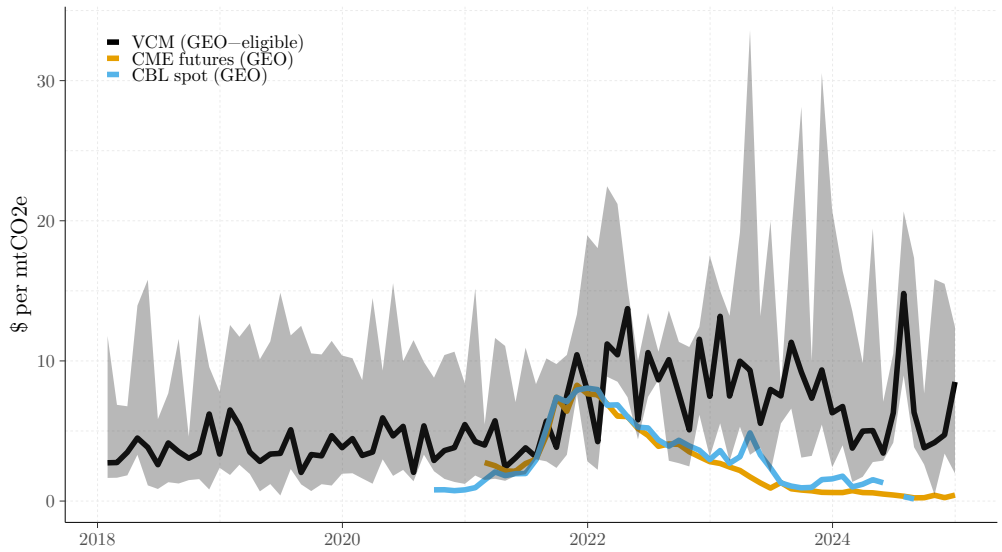


Figure 4: **CBL spot, CME futures, and VCM single-name credit transaction prices.** This figure plots the evolution of the median VCM single-name spot price, the CBL GEO spot price, and the CME GEO front-month futures price. Prices are in \$ per $mtCO_2e$. 'VCM' is the median monthly transaction price of GEO-eligible credits conditional on delivery within 10 days of purchase (a proxy for the spot price). The shaded area spans the 10th to 90th percentiles of spot VCM prices. The CBL and CME GEO contracts are baskets of CORSIA-eligible carbon credits. Figure A3 in the Appendix reports prices and volumes of other GEO-family contracts.

2021, and 2022, respectively.[27] They were soon followed by corresponding futures on the Chicago Mercantile Exchange (CME). GEO-family contracts are settled with the delivery of a single-name VCM credit that satisfies a set of eligibility rules, primarily those related to project technology and vintage. Therefore, basket products were intended to commoditize the VCM market. The price of the standardized credit basket should have served as an anchor for eligible single-name credits, like, for instance, the price of wheat from a given field links closely to a market price for the respective wheat grade based on measurable characteristics of the produce (protein content, moisture, kernel density, etc.), rendering the exact provenance less relevant (abstracting from transportation and storage costs).

[27] GEO (Global Emissions Offset) covers credits eligible under the aviation sector's Carbon Offsetting and Reduction Scheme for International Aviation (CORSIA), a global market-based scheme for addressing CO_2 emissions from international aviation. N-GEO (Nature-Based Global Emissions Offset) covers nature-based credits. C-GEO (Core Global Emissions Offset) covers non-nature-based credits that satisfy the Core Carbon Principles, developed by the Integrity Council for the Voluntary Carbon Market (an independent governance body established in 2021).

The commoditization attempt was short-lived: while basket prices initially followed single-name VCM prices, basket prices and trading volumes sharply declined within a few years. Figure 4 traces GEO spot and futures prices alongside single-name over-the-counter prices for eligible credits. After the basket product launch in 2020, CBL GEO prices followed the central tendency of single-name VCM prices for about a year.[28] From 2022, however, both spot and futures prices declined sharply relative to median single-name VCM prices, and by 2024, basket product prices and trading volumes were close to zero (Figure A3 in the Appendix reports prices and trading volumes of all GEO-family contracts, which follow similar patterns). Part of the GEO price decline may reflect a sharp expansion in the supply of basket-eligible credits starting in 2022.[29] However, because basket product trading volumes collapsed alongside prices, while eligible single-name credit prices were relatively more stable, the GEO family price drop is unlikely to be driven solely by market oversupply. Rather, a very broad definition of GEO credit baskets, coupled with buyers' willingness to pay for individual project characteristics as documented earlier in this paper, likely rendered standardized contracts redundant. The price of the cheapest-to-deliver basket-eligible single-name credit (which is linked to the standardized basket product price by no-arbitrage) is naturally a fraction of the median single-name credit price, as long as buyers pay non-standardized premia for project characteristics. If the goal of the basket product introduction was to create an actively traded commodity-like contract that anchors the price of a class of similar VCM credits, then this standardization attempt did not yield a relevant commodity market.

Despite a large gap in the average valuation of basket and over-the-counter VCM contracts, we find a link between the basket-market design and the levels of single-name credit prices. In

[28] One interpretation for this is that the basket market did behave like a commodity market early on. The other interpretation is that the early basket market presented ample arbitrage opportunities for those investors who could source cheap single-name credits and deliver against expensive spot or futures basket contracts.

[29] The CORSIA eligibility rules largely exclude credits from large-scale renewable energy and nature-based projects, as well as credits with vintages before 2016, generating an initially limited supply. Starting in 2022, issuance of eligible credits increased sharply. In 2024, issuance exceeded retirements by roughly a factor of three, and by year-end the stock of outstanding non-retired eligible credits stood at 81 million tons of CO_2 (see Figure A4).

	Dep. var.: log (Price)		
	(1)	(2)	(3)
Basket-eligible × Post-vintage-roll	0.23**	0.21*	0.18*
	(0.10)	(0.11)	(0.10)
Basket-eligible (dummy)	−0.03	−0.04	0.02
	(0.04)	(0.04)	(0.04)
Post-vintage-roll (dummy)	−0.34***	−0.66***	−0.25***
	(0.10)	(0.11)	(0.09)
Project FE	YES	YES	YES
Vintage Year FE	NO	YES	YES
Transaction Year FE	NO	NO	YES
Observations	7,203	7,203	7,203
Adjusted R^2	0.60	0.64	0.67

Table 12: **VCM credit prices and eligibility for delivery in basket products.**
The models in this table regress transaction prices in the dealer dataset on eligibility for delivery in spot and futures VCM basket products at CBL and CME. 'Basket eligible' takes the value of 1 for vintages eligible for delivery (six most recent year vintages as of July 1 of year T: $\{T-5,\ldots,T\}$), and 0 otherwise. 'Post-vintage-roll' takes the value of 1 for all contract signing dates after the aforementioned basket eligibility rule became effective (July 1, 2023), and 0 otherwise. Transaction-level controls include the year difference between the transaction date and the credit's vintage year (only in models 1 and 2), log volume, non-standard-currency (dummy), non-retirement delivery (dummy), and client home country log per-capita GDP. The standard errors (in parentheses) are clustered at the project level; $^{*}p<0.1$; $^{**}p<0.05$; $^{***}p<0.01$.

particular, CBL and CME regulate which credit vintage years are eligible for the basket-product settlement: the markets adopted rolling-vintage eligibility as of July 1, 2023 (pre-announced in April 2022). According to this rolling-vintage rule, only the six most recent calendar-year vintages are deliverable against CBL/CME N-GEO and C-GEO contracts, with the reset date on July 1 (i.e., on July 1, 2023, only vintages from 2018 to 2023 became eligible).[30] As Table 12 shows, such CBL/CME-basket-eligible credits (treated group) are traded at a premium of 20–26% after the rolling-vintage rule becomes effective (post-treatment).[31] The result controls for project, vintage year, and transaction year fixed effects. The estimated effect is also a lower bound of the demand-

[30] The CBL/CME rolling-vintage rule deviates from other credit eligibility rules, like aviation sector CORSIA standards, hence the CBL/CME vintage criteria was largely unforeseen, as evidenced by some investors' attempts to litigate the CME.

[31] Because the rule was announced in 2022 and became effective only on July 1, 2023, the post indicator captures the implementation response: to the extent that prices partly adjusted at announcement, the 20–26% premium is a lower bound on the total response. Figure A5 in the Appendix plots the loadings on the 'Basket eligible' dummy in Model (2) with 'Post-vintage-roll' dummy replaced by transaction year dummies. The plot illustrates that the effect is the strongest in 2024.

driven price jump in basket-eligible credits because an increase in the supply of eligible credits in recent years likely attenuates the treatment effect evaluated in Table 12.

The transaction data also suggest that the introduction of a publicly observed CBL/CME basket price benchmark was followed by a decrease in within-project price dispersion in the single-name market. The average price variation (the standard deviation of transaction prices in % of the average price) per project-year-quarter in a pooled sample of project-year-quarters with at least five trading days is about 34–36% pre-2021 (before the basket market) and 22% post-2021 at the reference category (renewables, projects with one buyer), as Models 1-4 in Table 13 suggest.[32] This effect does not hold under the model specification with project fixed effects (Model 5), suggesting that projects with lower price dispersion are overrepresented in the dataset post-2021. However, the contribution of additional buyers to within-project price variation declines post-CBL, even controlling for project identity. An additional unique client in a project-quarter is associated with about 1.6–2.6 p.p. higher coefficient of price variation pre-2021, but the effect drops by 1.1 p.p. post-2021 (Models 4 and 5 of Table 13). These empirical relations are consistent with theoretical predictions by Duffie et al. (2017), in which the introduction of a publicly observed benchmark in OTC markets limits dealers' ability to charge aggressive client-specific markups, which contribute to OTC price dispersion.[33]

Overall, we conclude that the basket product market, despite its failure to establish an actively traded commodity-like VCM price anchor, has shown some economic merit. It seems to have disciplined single-name VCM price dispersion by introducing a public price benchmark, and it also induced demand for basket-eligible credit vintages. Regarding VCM commoditization, one may

[32] Sample averages of the coefficient of price variation in this project-year-quarter sample are 44% and 22% pre- and post-2021, respectively.

[33] The introduction of the benchmark market coincided with other market developments that likewise contributed to VCM information transparency (e.g., the launch of AlliedOffsets data services). Thus, the reported effect of the benchmark market on price dispersion is likely an upper bound of what the market, per se, contributes.

	Dependent variable: $\frac{\sigma_p}{\bar{p}} \cdot 100$				
	(1)	(2)	(3)	(4)	(5)
Intercept	22.39^{***}	24.85^{***}	35.70^{***}	33.58^{***}	
	(3.91)	(4.02)	(6.78)	(6.76)	
No. unique buyers	1.90^{***}	2.08^{***}	1.97^{***}	2.65^{***}	1.58
	(0.61)	(0.64)	(0.68)	(0.81)	(1.25)
Expensive tech (dummy)		-8.14^{*}	-8.02^{**}	-13.23	
		(4.42)	(3.92)	(8.51)	
Post-CBL (dummy)			-14.64^{***}	-10.97^{*}	0.40
			(4.86)	(6.09)	(5.90)
No. unique buyers × Post-CBL				-1.10^{***}	-1.11^{***}
				(0.31)	(0.24)
Expensive tech × Post-CBL				7.02	
				(8.84)	
Project FEs	NO	NO	NO	NO	YES
Observations	337	337	337	337	337
Adjusted R^2	0.13	0.15	0.19	0.20	0.33

Table 13: **Factors of within-project price variation.**
The left-hand-side variable is the coefficient of variation of transaction prices—the standard deviation of price levels divided by the mean price, in percent ($\frac{\sigma_p}{\bar{p}} \cdot 100\%$)—computed per project per year-quarter. Only project-year-quarters with more than 5 trading days are included in the sample. There are 122 unique projects in this dataset. 'No. unique buyers' is the number of unique customers per project per year-quarter. 'Expensive tech' is the dummy that takes the value of 1 if the project is a nature-based (avoidance or removals) or a household one, and 0 otherwise. 'Post-CBL' is the dummy that takes the value of 1 for years 2021–2024, and 0 otherwise. The standard errors (in parentheses) are clustered at the project level; $^{*}p$ <0.1; $^{**}p$ <0.05; $^{***}p$ <0.01.

argue that as long as single-name prices are mainly driven by individual buyers' preferences, any basket product was destined to fail: a broadly defined basket quickly becomes worthless while a sufficiently narrow basket is virtually indistinguishable from a single-name contract.

5 Discussion

This section provides a structured discussion of the price patterns we document: we start from a stylized commodity benchmark and consider three potential departures from it. Appendix B formally develops the underlying pricing mechanisms.

The commodity benchmark. If carbon credits traded as a homogeneous commodity in a frictionless market between price-taking buyers and sellers, the law of one price would hold. A single market-clearing price, where the marginal abatement cost of supply equals the marginal buyer's willingness to pay for climate abatement, would prevail across all projects and buyers (Shapiro and Walker, 2025). This price would likely lie below the social cost of carbon: Since offsetting is voluntary, the marginal buyer may internalize only a fraction of the social cost of carbon, pushing the equilibrium price below it (see also Bellon et al., 2026) (see Appendix B.2). The patterns we document are inconsistent with this benchmark on three margins: cross-project dispersion of more than two orders of magnitude, substantial within-project price variation across buyers, and the price collapse of standardized exchange contracts.

Three channels could explain these patterns. Under *delivery risk*, promised carbon abatement may not materialize. Under *non-carbon preferences*, buyers value project features beyond carbon, with valuations that vary across buyers. Under *dealer price discrimination*, intermediaries charge different prices to different buyers for the same credit, exploiting buyers' limited ability to substitute across quotes.

Delivery risk. Credits may fail to deliver their promised abatement: a forestry project may burn, clean cookstoves may not be used, or a verification may be erroneous. If this is the only departure, more reliable projects should trade at a premium over less reliable ones, but two credits from the same project would still trade at a single price (see Appendix B.3).

Our data do not support this channel as the only departure from the commodity benchmark. Even though BeZero project reliability ratings correlate with prices in some specifications, the least-reliable technologies (nature-based avoidance and household) continue to trade at premia of 99–159% over renewables, while higher-rated waste management or industrial credits trade at no premium (Table 7). Delivery risk alone would generate the opposite ranking. The observed ranking could be explained by systematic biases in buyers' beliefs about technology-level reliability. However, this seems unlikely, as technology premia do not respond to the salient *New Yorker* quality signal of October 2023 (Table 9). Buyer-side sorting does not explain these patterns either: technology premia survive buyer fixed effects (Table 6, Model 9), so the same buyer pays more for nature-based than for renewable credits. Noisy reliability signals are also unlikely to generate the observed ranking: a classical lemons problem typically compresses the price–quality gradient rather than reverses it (Akerlof, 1970; Manshadi et al., 2025). Furthermore, cross-project variation in delivery risk cannot explain within-project dispersion.

Non-carbon preferences. Buyers may value project features beyond carbon abatement, such as biodiversity, local livelihoods, or emotionally salient narratives. When valuations of these features differ across buyers, willingness to pay becomes both project- and buyer-specific, and prices vary across projects according to the marginal buyer's tastes (see Appendix B.4).

This channel could explain the cross-project price dispersion we observe. Three patterns point to co-benefits as an important driver of cross-project price variation. First, our direct co-benefit

measure—the count of Sustainable Development Goals claimed in project documentation—is positively related to prices: an additional SDG is associated with a 4% premium (Table 8, Model 3). Second, Gold Standard certification, a registry whose methodology emphasizes co-benefits, commands a premium of up to 36% over CDM. Third, the largest premia accrue to nature-based and household projects, which the climate policy literature tends to associate with co-benefits (IPCC, 2022).

Dealer price discrimination. As in standard OTC markets (e.g., Duffie et al., 2005), VCM credits trade through specialized dealers who observe their counterparties, hold inventory, and negotiate bilaterally. Such markets can give rise to price discrimination across buyers. Pre-trade transparency is limited: most buyers may see only the quotes one or two dealers provide. Larger and more experienced buyers can compare quotes across dealers and against past trades; smaller and occasional buyers face the quote in front of them. The same credit may therefore trade at different prices to different buyers (see Appendix B.5).

This channel could explain the substantial within-project price dispersion we observe. Two pieces of evidence further support it. First, buyer-specific terms predict within-project prices: the 20 largest accounts pay 16–23% less than other buyers for the same project (Table 10), and credits delivered without immediate retirement trade at a 22–33% discount. Second, within-project price dispersion fell after CBL/CME basket products introduced a public price benchmark in 2021 (Table 13), our closest analog to a transparency shock.

In sum, no single departure from the commodity benchmark fits the observed pricing patterns on its own. These patterns point to a combination of heterogeneous buyer preferences over non-carbon project features and dealer price discrimination across buyers.

6 Conclusion

We show that the VCM does not price carbon as a commodity. Instead, prices exhibit extreme dispersion and reflect project, buyer, and individual-transaction characteristics rather than a common value for emission reductions. Buyers pay premia for project attributes beyond expected emission reductions. As a result, voluntary carbon market prices do not provide a market-based carbon price signal.

Our framework attributes VCM price patterns to two channels: buyer preferences over non-carbon project features and dealer price discrimination across buyers. The preference channel suggests VCM credits function more as differentiated certificates of voluntary corporate climate engagement, with benefits beyond climate mitigation, than as interchangeable units of carbon abatement. An important policy lever is therefore to align the climate claims firms can make based on carbon credit purchases with the actual climate impact of those credits. The price-discrimination channel suggests that the VCM's price-formation problem cannot be solved by existing exchange-listed standardized contracts, which, as we show, have largely failed. The VCM market still requires pre-trade transparency in the form of meaningful public price benchmarks to limit dealers' market power. Together, these levers would preserve the VCM's appeal as a Coasian instrument that mobilizes private capital across jurisdictions, while correcting the specific failures of its current price discovery.

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Appendix A Additional Figures and Tables

	Dependent variable: log (Price)								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Intercept	0.77***	1.20***	1.35***	1.50***	1.32***	1.69***			
	(0.06)	(0.14)	(0.48)	(0.49)	(0.49)	(0.48)			
Tech: Industrial	0.40***	0.22	0.22	0.12	0.09	0.12	−0.04	−0.19	−0.30***
	(0.11)	(0.14)	(0.14)	(0.15)	(0.15)	(0.13)	(0.08)	(0.15)	(0.11)
Tech: NB Removal	1.35***	1.38***	1.38***	1.12***	1.08***	1.08***	1.13***	0.91***	0.91***
	(0.25)	(0.25)	(0.25)	(0.32)	(0.31)	(0.29)	(0.29)	(0.15)	(0.10)
Tech: NB Avoidance	1.00***	1.05***	1.04***	0.83***	0.83***	0.85***	0.82***	0.51***	0.57***
	(0.10)	(0.10)	(0.10)	(0.12)	(0.12)	(0.12)	(0.10)	(0.11)	(0.08)
Tech: Household	1.23***	0.89***	0.86***	0.62***	0.63***	0.54***	0.54***	0.08	0.03
	(0.07)	(0.10)	(0.12)	(0.13)	(0.14)	(0.13)	(0.11)	(0.13)	(0.13)
Tech: Waste	0.68***	0.49***	0.50***	0.42**	0.38*	0.44**	0.46***	0.30**	0.28***
	(0.13)	(0.18)	(0.19)	(0.19)	(0.20)	(0.19)	(0.16)	(0.14)	(0.10)
Registry: Gold Standard		−0.04	−0.06	−0.19	−0.21	−0.08	−0.32**	0.52**	0.82***
		(0.16)	(0.18)	(0.18)	(0.19)	(0.18)	(0.13)	(0.25)	(0.18)
Registry: Verra		−0.49***	−0.51***	−0.56***	−0.60***	−0.44***	−0.61***	0.26	0.46***
		(0.12)	(0.15)	(0.16)	(0.17)	(0.16)	(0.12)	(0.23)	(0.16)
log (GDP p.c.)			−0.01	0.05	0.08	0.09**	0.04	5.66***	3.90***
			(0.05)	(0.05)	(0.05)	(0.05)	(0.04)	(1.22)	(1.11)
log (Country emissions)				−0.09***	−0.10***	−0.09***	−0.06***	0.21	−2.43***
				(0.02)	(0.02)	(0.02)	(0.02)	(1.04)	(0.70)
Project age (years)					0.01	0.002	−0.01	−0.01	−0.01*
					(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
Project size (std. per tech)					0.07**	0.08**	0.05	−0.05*	0.01
					(0.03)	(0.03)	(0.03)	(0.03)	(0.02)
log (Trade volume)						−0.05***	−0.06***	0.01	−0.04***
						(0.01)	(0.01)	(0.01)	(0.01)
Credit vintage (years)						−0.05***	−0.07***	−0.05***	−0.07***
						(0.01)	(0.01)	(0.01)	(0.01)
Year FEs	NO	NO	NO	NO	NO	NO	YES	NO	YES
Country FEs	NO	NO	NO	NO	NO	NO	NO	YES	YES
Observations	3,875	3,875	3,875	3,875	3,875	3,875	3,875	3,875	3,875
Adjusted R^2	0.25	0.27	0.27	0.30	0.30	0.32	0.63	0.45	0.72

Table A1: **Models for transaction log-prices and project characteristics in the CBL+Platts sample.**
This table is analogous to Table 6, but for the CBL+Platts sample. The dependent variable is the natural logarithm of the transaction price in the CBL+Platts sample. Technology and registry indicators correspond to project technologies and VCM credit registries. The reference (dropped) categories are, respectively, renewable energy projects (tech) and a collective of registries beyond Verra and GS (registry). Annual GDP per capita (in constant 2021 USD) and country emissions are sourced from the World Bank WDI dataset. Project age is the difference in years between the traded certificate's vintage year and the project launch (the earliest available vintage per project). Project size is the total emission reduction expected at project launch, standardized to zero mean and unit variance per project technology. The carbon credit vintage is the year difference between the transaction date and the credit's vintage year. The standard errors (in parentheses) are clustered at the project level; $^{*}p<0.1$; $^{**}p<0.05$; $^{***}p<0.01$.

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	2019	2020	2021	2022	2023	2024	Total
No. transactions	154	443	1347	977	649	305	3875
By project technology, % of total							
Renewables	77.9	77.9	64.3	56.5	34.4	31.1	56.8
Industrial	1.3	1.1	2.4	5.3	12.8	10.2	5.3
NB Removals	2.6	1.1	4.2	3.8	6.9	9.5	4.6
NB Avoidance	15.6	19.0	22.4	31.0	37.6	39.7	27.8
Household	0.6	0.5	3.3	1.9	6.9	5.6	3.3
Waste	1.9	0.5	3.3	1.4	1.4	3.9	2.2

Table A2: **No. transactions, by year and project technology, in the CBL+Platts sample.** This table is analogous to Table 1(B), but for the CBL+Platts sample. Project technology classification is the same as in the core VCM dealer-to-customer transactions sample.

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	Dependent variable: log (Price)								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Intercept	0.89***	0.94***	0.12	0.12	0.39				
	(0.07)	(0.11)	(0.41)	(0.41)	(0.40)				
Tech: Industrial	−0.26	−0.29	−0.30	−0.33	−0.30	−0.05	−0.09	−0.01	−0.04
	(0.26)	(0.28)	(0.24)	(0.25)	(0.29)	(0.20)	(0.17)	(0.20)	(0.16)
Tech: NB Removal	1.42***	1.46***	1.38***	1.37***	1.38***	1.43***	1.71***	1.73***	1.86***
	(0.12)	(0.12)	(0.15)	(0.14)	(0.14)	(0.13)	(0.18)	(0.21)	(0.21)
Tech: NB Avoidance	0.71***	0.77***	0.72***	0.73***	0.72***	0.80***	0.63***	0.35	0.80***
	(0.11)	(0.12)	(0.14)	(0.14)	(0.14)	(0.11)	(0.19)	(0.25)	(0.23)
Tech: Household	0.87***	0.82***	0.83***	0.86***	0.79***	0.79***	1.02***	1.06***	1.10***
	(0.12)	(0.13)	(0.15)	(0.15)	(0.16)	(0.11)	(0.10)	(0.11)	(0.14)
Tech: Waste	0.18*	0.19*	0.17*	0.18*	0.16*	0.23***	0.23***	0.05	0.12
	(0.10)	(0.10)	(0.10)	(0.10)	(0.09)	(0.08)	(0.08)	(0.09)	(0.10)
Registry: Gold Standard		0.02	−0.004	−0.01	−0.07	−0.07	0.31***	0.16	0.12
		(0.14)	(0.14)	(0.14)	(0.13)	(0.10)	(0.11)	(0.14)	(0.13)
Registry: Verra		−0.11	−0.12	−0.13	−0.10	−0.23**	0.02	−0.16	−0.12
		(0.13)	(0.13)	(0.13)	(0.12)	(0.10)	(0.10)	(0.14)	(0.15)
log (GDP p.c.)			0.13**	0.12**	0.12**	0.03	0.16	−0.33	−0.18
			(0.06)	(0.06)	(0.06)	(0.04)	(0.87)	(0.89)	(0.82)
log (Country emissions)			−0.05**	−0.04*	−0.04*	−0.03**	0.54	0.37	−0.25
			(0.02)	(0.02)	(0.02)	(0.01)	(0.47)	(0.60)	(0.45)
Project age (years)				0.02*	0.02	0.01	0.004	−0.002	−0.002
				(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
Project size (std. per tech)				−0.04	−0.03	−0.07**	−0.08***	−0.08***	−0.08***
				(0.05)	(0.05)	(0.04)	(0.03)	(0.03)	(0.03)
log (Trade volume)					−0.08***	−0.11***	−0.09***	−0.08***	−0.05***
					(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
Credit vintage (years)					−0.03**	−0.06***	−0.04***	−0.03***	−0.03***
					(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
Non-standard currency (dummy)					0.58**	0.32***	0.30***	0.37**	−0.85***
					(0.27)	(0.10)	(0.09)	(0.15)	(0.18)
Delivery delay (months)					0.01	−0.01	−0.01	−0.01	−0.01
					(0.02)	(0.01)	(0.01)	(0.01)	(0.01)
Year FEs	NO	NO	NO	NO	NO	YES	YES	YES	YES
Country FEs	NO	NO	NO	NO	NO	NO	YES	YES	YES
Clients with 10+ trades	NO	NO	NO	NO	NO	NO	NO	YES	YES
Client FEs	NO	NO	NO	NO	NO	NO	NO	NO	YES
Observations	1,564	1,564	1,564	1,564	1,564	1,564	1,564	1,054	1,054
Adjusted R^2	0.33	0.33	0.35	0.35	0.39	0.63	0.71	0.76	0.84

Table A3: **Models for VCM prices and project characteristics (non-retired credits only).** This table is analogous to Table 6, but the models are estimated in the subsample of transactions where the credit was transferred to the buyer without immediate retirement by the dealer on behalf of the client. The dependent variable is the natural logarithm of the transaction price. Technology and registry indicators correspond to the categories in Table 3. The reference (dropped) categories are, respectively, renewable energy projects (tech) and the CDM (registry). Annual GDP per capita (in constant 2021 USD) and country emissions are sourced from the World Bank WDI dataset. Project age is the difference in years between the traded certificate's vintage year and the project launch (the earliest available vintage per project). Project size is the total emission reduction expected at project launch, standardized to zero mean and unit variance per project technology. 'Top-tier client' takes the value 1 if the customer is among the top-20 trading accounts by transaction volume across the entire dataset. The carbon credit vintage is the year difference between the transaction date and the credit's vintage year. The non-standard currency indicator identifies transactions settled in a currency other than EUR, USD, GBP, AUD, or CHF. Delivery delay is the difference in months between the signing and delivery dates. Models 8 and 9 are estimated using the subsample of customers with at least 10 transactions. The standard errors (in parentheses) are clustered at the project level; $^{*}p<0.1$; $^{**}p<0.05$; $^{***}p<0.01$.

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	Dep. var.: log (Price)		
	(1)	(2)	(3)
NB Avoidance (dummy)	0.50***	0.55***	0.36***
	(0.09)	(0.09)	(0.08)
Post Guardian (dummy)		0.21*	−0.001
		(0.11)	(0.12)
NB Avoidance × Post Guardian		−0.19	−0.02
		(0.14)	(0.08)
Year-quarter FE	YES	YES	YES
Client FE	NO	NO	YES
Observations	7,203	7,203	7,203
Adjusted R^2	0.34	0.34	0.69

Table A4: **Nature-based avoidance price premia around a public quality controversy (Guardian article).**
This table estimates how the price premium of nature-based avoidance projects, relative to all other projects in the dealer-to-customer sample, changes after the January 18, 2023, publication of a *Guardian* article documenting quality concerns about nature-based avoidance projects in Latin America. 'Post Guardian' equals one after January 18, 2023, and zero otherwise. All models include calendar-year-quarter fixed effects. There are 18 and 36 transactions in Q1 2023 in the 'NB Avoidance' group pre- and post-Guardian article publication, respectively. Control variables are registry dummies, transaction volume, vintage, non-standard delivery currency, and non-retirement delivery; $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

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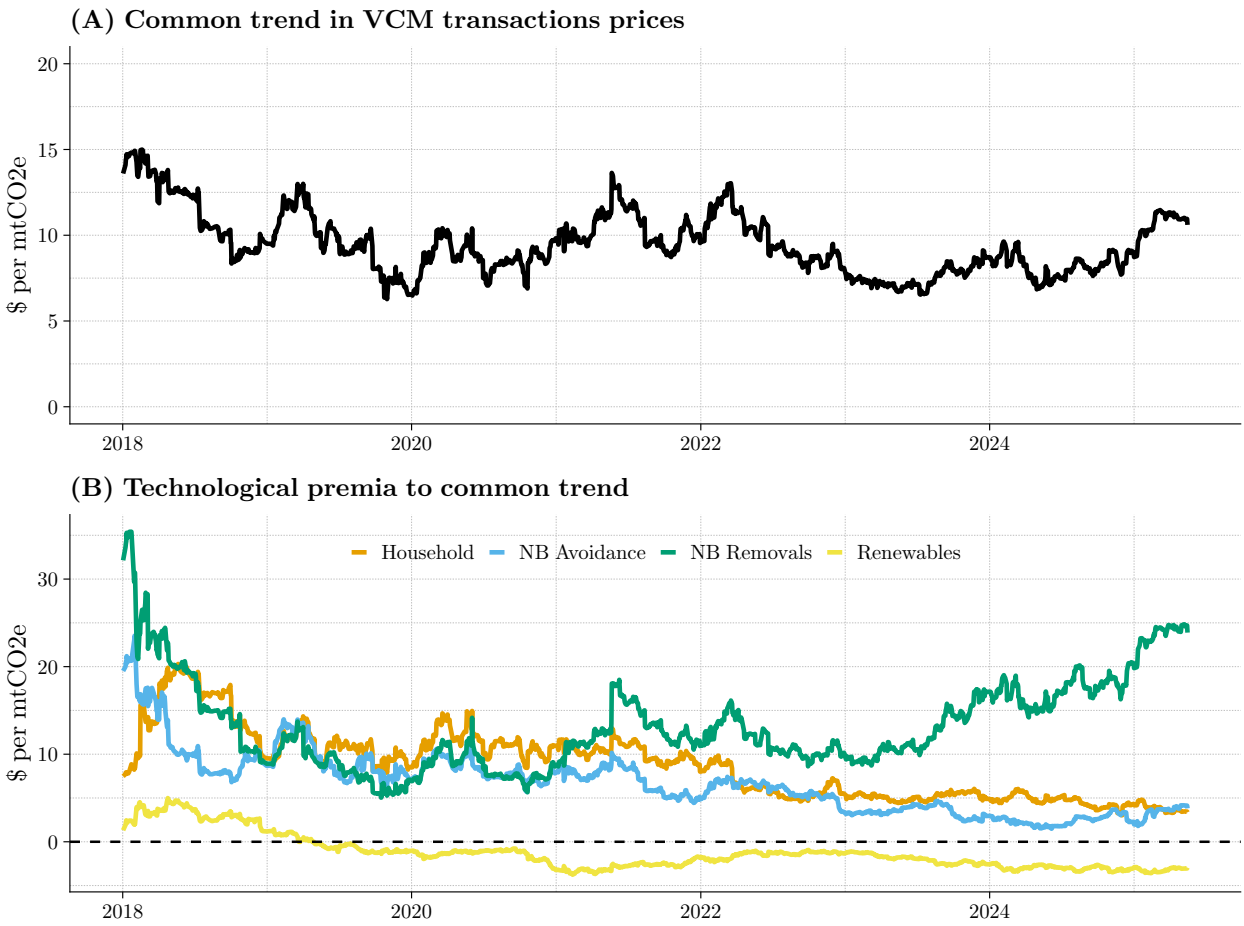


Figure A1: **Common VCM price trend and technology premia in a dynamic model.** The common trend and technology premia are estimated in a dynamic linear state-space model in the spirit of Francke and Vos (2004), applied to VCM log-prices. In addition to technology premia, the model includes geographical and registry trends. The common trend and all premia are assumed to follow random walks in logs. The model is estimated at the project-calendar-day level. The log-likelihood is evaluated using a modified Kalman filter that skips the update step when no transaction is observed for a project on a given day. Diffuse initialization is used. State-transition and measurement-error variances are estimated by maximizing the log-likelihood using a generalized simulated-annealing algorithm. The estimates shown are from a filter-smoother run at the optimized parameters and are rescaled from log-prices to dollars per $mtCO_2e$.

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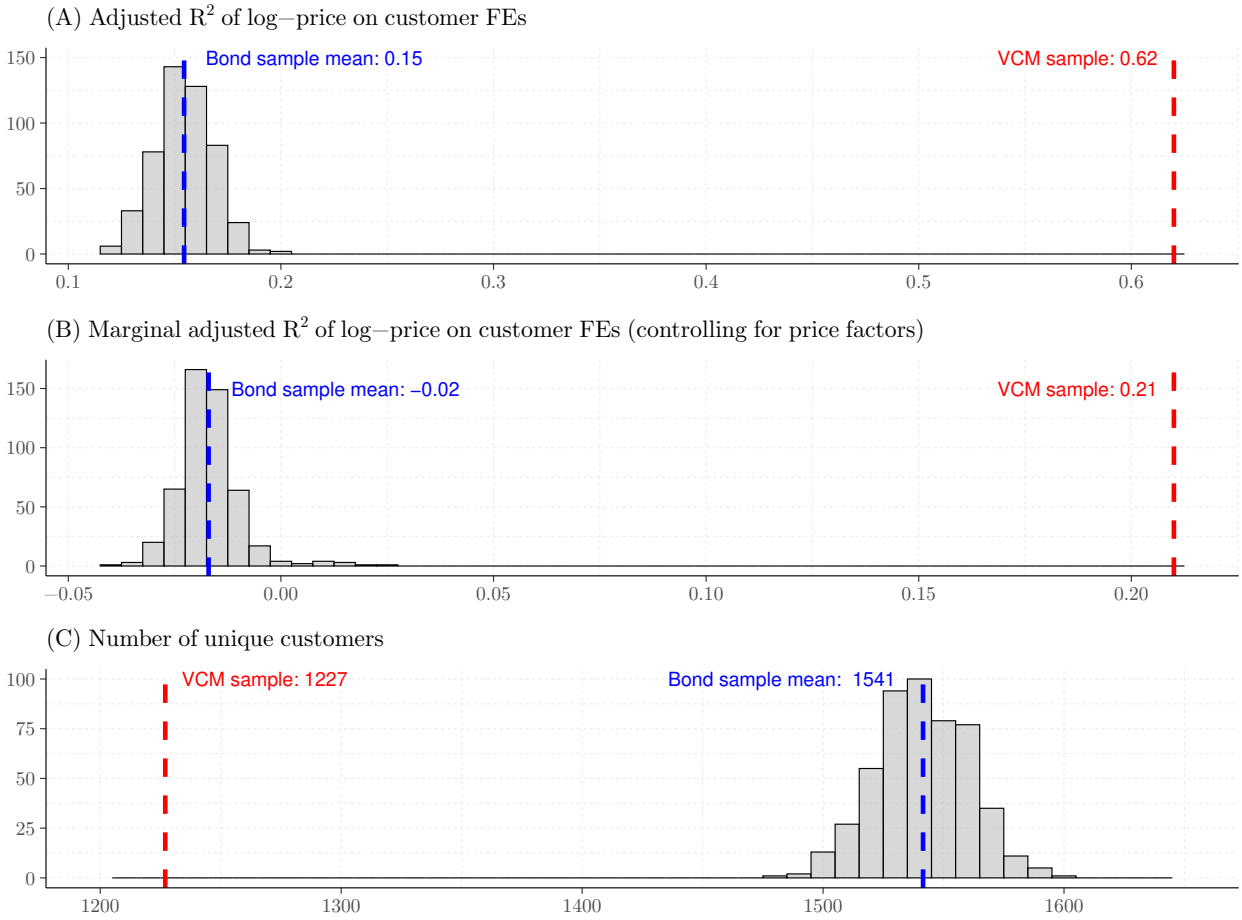


Figure A2: **Log-price variance explained by customer-ID fixed effects in the VCM and corporate bond samples.**
The bond transaction tape is from the National Association of Insurance Commissioners (NAIC) and comprises (after usual filters) U.S. corporate bond trades between bond dealers and insurers (customers) in 2017–2023. From this bond tape, we draw 500 times a random sample of transactions of the same size as our VCM sample (7.2k obs.). In each corporate bond sample, we regress the log of the transaction price, first, on customer IDs (fixed effects) only, then on relevant bond price factors (sum of promised cash flows discounted at the Treasury yield curve as of transaction date, supplemented with a rich set of fixed effects related to transaction time, bond coupon, rating, maturity, and issue size), and finally on both price factors *and* customer IDs. We plot the distribution of adjusted R^2 for customer-ID-only regression in panel (A); the difference in adjusted R^2 between price-factor models with and without customer IDs – in panel (B). Customer fixed effects in the bond samples are at the fund/portfolio rather than the insurer holding company level. The distribution of the number of unique customers across sample draws is in panel (C). In all panels, blue dashed lines represent the means across bond sample draws. The red lines are the reference VCM sample values from Table 11.

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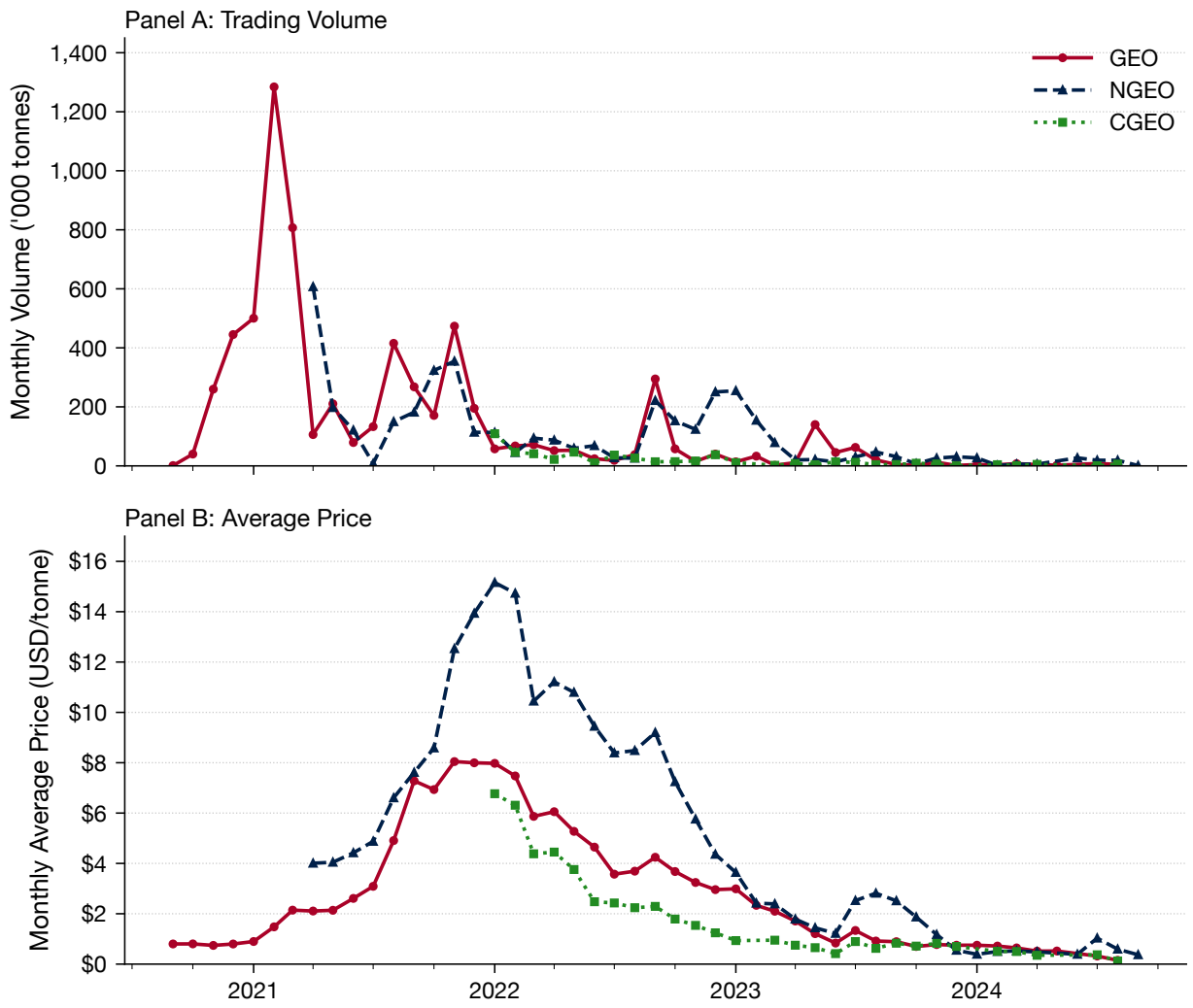


Figure A3: **Time series of basket trading volumes.**
Panel A shows monthly total trading volume in thousands of tonnes CO_2e. Panel B shows the monthly volume-weighted average transaction price in USD per tonne CO_2e. Both panels cover non-negotiated trades on the CBL exchange. GEO = Global Emissions Offset; NGEO = Nature-based Global Emissions Offset; CGEO = Core Global Emissions Offset. Sample period: September 2020 to September 2024.

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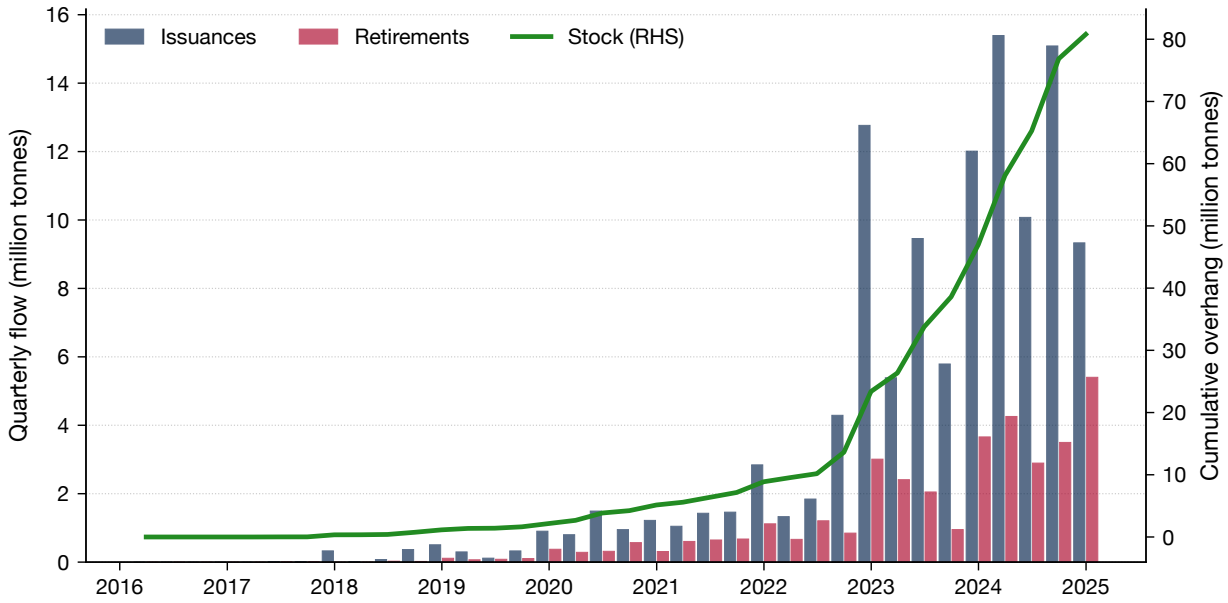


Figure A4: **Issuances, retirements, and stock of CORSIA-eligible projects.**
Bars show quarterly credit issuances (dark blue) and retirements (dark red) in million tonnes CO_2e (left axis). The solid green line shows the cumulative stock of eligible credits, defined as cumulative issuances minus cumulative retirements (right axis). The sample covers Verra (VCS) and Gold Standard (GSR) projects that ever achieved CORSIA eligibility, restricted to vintage year ≥ 2016.

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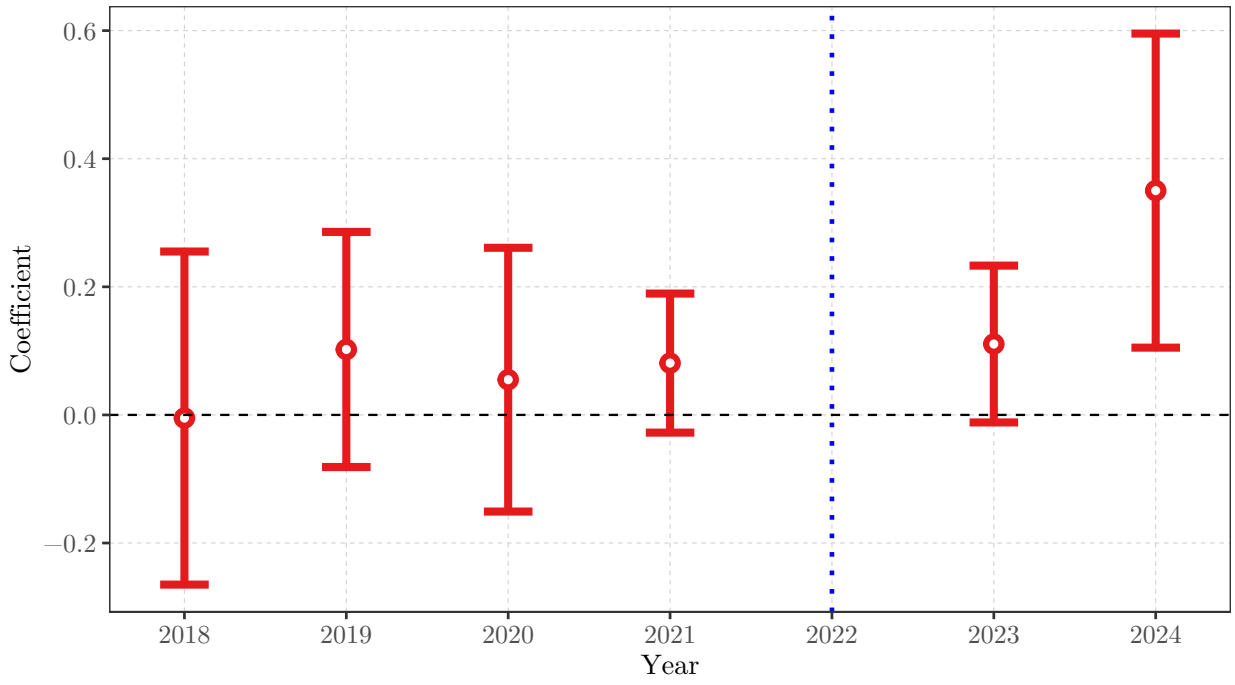


Figure A5: **VCM credit prices and eligibility for delivery in basket products.** This figure plots the loading on the Basket-eligible dummy in the model similar to model (2) of Table 12, but with the 'Post-vintage-roll' dummy replaced by transaction year dummies. The reference year is 2022; the vertical axis shows the log-difference in the price of basket-eligible vs. non-eligible credits relative to the 2022 difference. The bars are 90% confidence bands (standard errors clustered at the project level).

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Appendix B A simple framework for VCM pricing

This appendix develops a stylized partial-equilibrium framework for VCM pricing. We start from a benchmark in which carbon credits trade as a homogeneous commodity priced by carbon abatement alone, relax three of its assumptions one at a time, and combine the three relaxations into a single pricing equation.

B.1 Setup

Consider N firms with emissions $e_i > 0$. Firm i internalizes a fraction $\beta_i \in [0, 1]$ of the social cost of its emissions—for example, because of stakeholder pressure, anticipated regulation, or intrinsic pro-social preferences (see Broccardo et al., 2022, for a formal treatment of pro-social preferences in the context of ESG). We write this internalized cost as $\beta_i \cdot s$ per ton, where $s > 0$ is the social cost of carbon, formally the net present value of the marginal disutility from one additional ton of atmospheric CO_2e.

On the supply side, J project developers issue carbon credits—units representing one ton of CO_2e reduction or removal—at heterogeneous marginal costs $c_j > 0$ up to a finite issuance cap $\bar{q}_j$, where j indexes projects. We work with a continuum-of-projects approximation, under which aggregate supply is continuous in price. Throughout, equilibrium is competitive market clearing under price taking, except in Section B.5, where intermediation creates local market power.

B.2 Pricing under the commodity benchmark

We first characterize equilibrium under three benchmark conditions, listed in the order in which they are relaxed below:

(i) *No delivery risk.* Every credit delivers the promised carbon abatement with certainty.

(ii) *Homogeneous product.* Buyers value only the carbon abatement of a credit; one ton of abated CO_2e from one project is a perfect substitute for one ton from another project.

(iii) *Frictionless trading.* Buyers and sellers are price takers and can trade without search or bargaining frictions.

Under (i)–(iii), each credit delivers one ton of CO_2e and buyers care only about carbon abatement. Offsetting one ton therefore lowers firm i's internalized carbon cost by $\beta_i \cdot s$, so its marginal willingness to pay is

$$v_i \;=\; \beta_i \cdot s. \tag{3}$$

Aggregate demand $D(p) = \sum_{i:\, v_i \geq p} e_i$ is decreasing in p. Together with an upward-sloping supply curve—projects ordered by ascending c_j—it yields a market-clearing price p^* satisfying $\min_j\{c_j\} \leq p^* \leq \max_i\{v_i\}$, generically unique under the continuum approximation. In an interior competitive equilibrium, p^* also equals the marginal cost of the highest-cost project supplied.

Proposition 1 (Commodity-benchmark pricing)**.** *Under conditions (i)–(iii), all credits trade at a single price*

$$p^* \;=\; \beta^* \cdot s, \tag{4}$$

where $\beta^* = \min\{\beta_i : i \text{ trades}\}$ is the internalization parameter of the marginal buyer. Because participation is voluntary, $\beta^* \leq 1$, so $p^* \leq s$.

Prices strictly below the social cost of carbon are therefore expected in the VCM, not anomalous (see also Bellon et al., 2026). All credits trade at the same price regardless of project type, host country, or buyer identity. We refer to (i)–(iii) plus equations (3)–(4) as the *commodity benchmark*.

B.3 Relaxing (i): delivery risk

We relax (i) by allowing credits to differ in delivery reliability while maintaining (ii)–(iii). Let $q_j \in (0, 1]$ denote the expected carbon abatement per credit from project j, that is, the probability-weighted fraction of one ton of CO_2e that the credit delivers. In practice, a credit may fall short of its promised abatement for a variety of reasons: a forestry project may burn, a verification may turn out to be erroneous, or carbon stored in soil or biomass may be re-emitted. The scalar q_j summarizes these contingencies into a single expected value, which suffices for the sign restriction below.[34] We initially assume buyers observe q_j. A credit from project j now reduces firm i's internalized carbon cost by $\beta_i \cdot s \cdot q_j$, so willingness to pay becomes

$$w_{ij} \;=\; \beta_i \cdot s \cdot q_j. \tag{5}$$

Proposition 2 (Pricing under delivery risk)**.** *Under conditions (ii)–(iii), with buyers observing q_j, and equation (5), the equilibrium price of any traded project j satisfies*

$$p_j \;=\; \beta^* \cdot s \cdot q_j, \tag{6}$$

where β^* is the internalization parameter of the marginal buyer of expected carbon abatement. In particular, $p_j/p_k = q_j/q_k$ for any two traded projects.

The argument is immediate. Because $w_{ij} = \beta_i s q_j$ is multiplicatively separable in i and j, willingness to pay per unit of *expected carbon abatement*, $w_{ij}/q_j = \beta_i s$, does not depend on j. Once credits are converted into expected-ton units, they are perfect substitutes, so a single price clears the market for expected abatement. Delivery risk alone therefore preserves the law of one price in the relevant units: cross-project prices vary only in proportion to q_j.

Quality information and beliefs. Proposition 2 assumes buyers observe q_j. In practice, beliefs are imprecise and potentially heterogeneous, but the sign discipline survives. When beliefs are common across buyers, projects perceived as delivering more carbon abatement command higher prices regardless of belief precision; with a coarse project-type signal $\tau(j)$, credits within a given type trade at a price proportional to $E[q_j \mid \tau(j)]$. When beliefs differ across buyers, willingness to pay becomes buyer-project-specific, but under competitive pricing (iii) this affects which buyer is marginal for each project rather than generating within-project dispersion. With private quality information, a classical lemons problem (Akerlof, 1970; Manshadi et al., 2025) compresses the price–quality gradient rather than reversing it.

B.4 Relaxing (ii): non-carbon valuations

We next relax (ii) while maintaining (i) and (iii). Project j carries a vector of non-carbon attributes $\mathbf{x}_j \in \mathbb{R}^K$ in addition to its carbon abatement. Buyer i values these attributes at $\boldsymbol{\theta}_i' \mathbf{x}_j$, where $\boldsymbol{\theta}_i \in \mathbb{R}^K$ captures its marginal valuation of each attribute and is heterogeneous across buyers. The motives

[34] We follow Kim et al. (2024) in using q_j as the offset quality parameter; in a compliance-market context, Bellon et al. (2026) parameterize the same delivery friction through a complementary invalidation probability (q and $1 - \rho$ are interchangeable).

behind $\boldsymbol{\theta}_i$ may be intrinsic (co-benefits such as biodiversity or local livelihoods, warm-glow from emotionally salient projects, Andreoni 1990) or instrumental (regulatory anticipation, stakeholder signaling); the framework treats them as observationally equivalent. Willingness to pay becomes

$$w_{ij} \;=\; \beta_i \cdot s \;+\; \boldsymbol{\theta}_i' \mathbf{x}_j. \tag{7}$$

Proposition 3 (Pricing under non-carbon valuations)**.** *Under conditions (i) and (iii) and equation (7), projects with distinct attribute vectors are imperfect substitutes, and the market for each project j clears at*

$$p_j \;=\; \beta_j^* \cdot s \;+\; (\boldsymbol{\theta}_j^*)' \mathbf{x}_j, \tag{8}$$

where $(\beta_j^*, \boldsymbol{\theta}_j^*)$ is the marginal buyer of project j, that is, the lowest-WTP buyer among those who purchase a credit from project j in equilibrium. Within a project, all credits trade at the single price p_j.

The result follows from frictionless competitive trading: with J differentiated goods, each project market clears at the WTP of the project's marginal buyer. Sorting of buyers across projects according to $\boldsymbol{\theta}_i$ generates differences in $(\beta_j^*, \boldsymbol{\theta}_j^*)$ and hence in p_j. The within-project law of one price still holds because (iii) holds.

B.5 Relaxing (iii): search frictions and intermediation

We finally relax (iii) while maintaining (i) and (ii). Specialized intermediaries broker most OTC VCM trades. They observe their client books and maintain relationships with both project developers and buyers, while buyers typically see only the offers they receive from one or two intermediaries. We capture this asymmetry with a bilateral-matching protocol in the spirit of Duffie et al. (2005). Intermediaries hold inventory of homogeneous credits at an effective per-unit cost equal to the competitive benchmark price p^0 that would prevail absent buyer-specific search frictions, which also functions as their reservation price in bilateral negotiation. Each buyer i is matched with one intermediary, who observes the buyer's reservation value and posts a take-it-or-leave-it offer. With probability $\kappa_i \in [0,1]$, the buyer obtains a competing quote and Bertrand competition drives the retail price to p^0; with probability $1-\kappa_i$, the buyer is captive, and the matched intermediary extracts the full reservation value. Trade occurs whenever the buyer's reservation value is at least p^0. The parameter κ_i summarizes the buyer's search capacity: size, market experience, and procurement resources.

Under (i)–(ii), willingness to pay is the commodity-benchmark value $w_i = \beta_i \cdot s$.

Proposition 4 (Pricing under search frictions)**.** *Under conditions (i)–(ii) and the bilateral-matching protocol above, the expected transaction price faced by buyer i is*

$$p_i \;=\; (1-\kappa_i)\,\beta_i \cdot s \;+\; \kappa_i\, p^0, \tag{9}$$

where p^0 is the frictionless market-clearing price, i.e., the price that solves $S(p^0) = \sum_{i:\,\beta_i s \geq p^0} e_i$ evaluated at $\kappa_i = 1$ for all i. Existence and uniqueness follow from the usual monotonicity of supply and residual demand.

Because credits are homogeneous under (ii), buyers transact at p^0 in the competitive limit ($\kappa_i = 1$ for all i) and at their reservation values when captive ($\kappa_i = 0$). Search frictions thus generate

cross-buyer price dispersion within the (single) project market while leaving the law of one price intact at the competitive benchmark.[35]

B.6 Combined pricing

Combining all three relaxations, buyer i's willingness to pay for a credit from project j is

$$w_{ij} \;=\; \underbrace{\beta_i \cdot s \cdot q_j}_{\text{carbon value}} \;+\; \underbrace{\boldsymbol{\theta}_i' \mathbf{x}_j}_{\text{non-carbon value}}, \tag{10}$$

which nests equation (3) at $q_j = 1$ and $\boldsymbol{\theta}_i = \mathbf{0}$, equation (5) at $\boldsymbol{\theta}_i = \mathbf{0}$, and equation (7) at $q_j = 1$. The transaction price faced by buyer i in project j follows from the bilateral protocol of Section B.5, now applied project by project because projects are imperfect substitutes once $\boldsymbol{\theta}_i \neq \mathbf{0}$.

Proposition 5 (Combined pricing)**.** *Under equation (10) and the bilateral-matching protocol of Section B.5, the expected transaction price for buyer i in project j is*

$$p_{ij} \;=\; (1 - \kappa_i)\, w_{ij} \;+\; \kappa_i\, p_j^0, \tag{11}$$

where the project-level benchmark price p_j^0 is the frictionless market-clearing price for project j. Formally, if buyer i demands d_{ij} units when participating, p_j^0 solves $S_j(p_j^0) = \int d_{ij}\, \mathbf{1}\{w_{ij} \geq p_j^0\}\, di$, evaluated at $\kappa_i = 1$ for all i. Existence and uniqueness follow from the usual monotonicity of supply and residual demand.

Equation (11) nests Propositions 1–4 as special cases: setting $\kappa_i = 1$ for all i collapses p_{ij} to the frictionless benchmark p_j^0, in which $\boldsymbol{\theta}_i = \mathbf{0}$ further yields the carbon-only pricing of Propositions 1–2, and $q_j = 1$ yields the non-carbon-valuation pricing of Proposition 3. When trade occurs, the protocol bounds $p_{ij} \in [p_j^0,\, w_{ij}]$. The three primitives q_j, $\boldsymbol{\theta}_i$, and κ_i act through distinct moments: q_j shifts the carbon-value component of w_{ij} and is project-specific; $\boldsymbol{\theta}_i$ generates buyer-by-project variation in w_{ij} through the interaction $\boldsymbol{\theta}_i' \mathbf{x}_j$; κ_i governs the buyer's location on $[p_j^0, w_{ij}]$ and is project-independent.

Mapping to the basic pricing facts. The three primitives q_j, $\boldsymbol{\theta}_i$, and κ_i map onto the three pricing facts documented in Sections 4.1 and 4.2. *Between-project price dispersion* requires heterogeneity in q_j (Proposition 2) or in $\mathbf{x}_j$ under heterogeneous $\boldsymbol{\theta}_i$ (Proposition 3): at $q_j = 1$ and $\boldsymbol{\theta}_i = \mathbf{0}$ for all i, all credits trade at the same price. *Within-project price dispersion across buyers* requires $\kappa_i < 1$ for some buyers (Proposition 4); setting $\kappa_i = 1$ for all buyers collapses p_{ij} to the frictionless benchmark p_j^0, so neither q_j nor $\boldsymbol{\theta}_i$ alone produces within-project dispersion. *The decoupling of technology premia from external carbon-delivery measures* requires $\boldsymbol{\theta}_i' \mathbf{x}_j$ to dominate the carbon component for technologies with low external q_j: under Proposition 2 alone, prices would order monotonically in q_j, so the observed ranking, in which lower-rated technologies trade at higher prices than higher-rated ones, requires non-carbon valuations correlated with low q_j at the technology level. Matching the three facts simultaneously requires both non-carbon valuations and search frictions; delivery risk alone is consistent with neither the within-project dispersion nor the observed price–quality ordering.

[35] The closed form depends on the take-it-or-leave-it structure, but the qualitative predictions are robust to alternative bargaining protocols. Under Nash bargaining over the surplus $\beta_i s - p^0$ with intermediary bargaining weight $1 - \kappa_i$, the price is $p_i = p^0 + (1 - \kappa_i)(\beta_i s - p^0)$, which delivers the same comparative statics.